Know Thyself: Free Credit Reports and the Retail Mortgage Market *

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Abstract

Credit history is the top reason for mortgage rejection in the U.S. Exploiting the 2004 federal extension of existing state laws allowing free credit reports, I document that a reduction in consumers' cost of accessing their own credit history information leads to an increase in mortgage demand, mortgage approval ratio and first-time homebuyers. Also, consumers become less likely to default on the mortgages, even during and after the 2008 financial crisis. Further, the mortgages become costlier, suggesting that a demand-pull effect is more instrumental in driving the higher mortgage approvals and origination, than a supply-push effect is. Thus, I conclude that the higher mortgage approval ratio is due to higher quality applicants in the market. Pool improvement occurs mainly among prime and more educated population, and among the bottom income quartile consumers. Overall, results suggest that costly credit history information excludes some creditworthy consumers from the credit market.

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1 Introduction

A salient feature of rejections in the retail mortgage market in the US is that most rejections are due to consumers' own credit history. Between 2000 and 2008, 28% of all rejections, or 39.4% of all rejections which mentioned a denial reason was due to consumers' credit history.[1] Even if we attribute some of these rejections to mistakes by lenders, the question arises, why numerous consumers apply for mortgage, only to be rejected subsequently for their own credit history, while bearing the direct and indirect cost of the rejection? This question becomes even more intriguing by the fact that many consumers suppress their demand for credit, because they anticipate that they will be rejected. Specifically, about 15% of the US population wishes to, but does not apply for credit (Survey of Consumer Finances, SCF).[2] While many of these consumers would be correct in their assessment that they would be rejected, many may be overestimating the probability of rejection, and thus, suppressing their need for credit despite being creditworthy. Overall, these two facts suggest that consumers are imperfectly informed about their own creditworthiness.[3] This begs the question, does the economic cost of accessing one's own credit history play a role in consumer's decision to participate in the credit market? How does this cost affect the credit demand and approval ratio?

The goal of this paper is to examine the link between the economic cost of accessing one's own credit history information and the mortgage approval ratio and demand for mortgage. I use a difference-in-differences (DID) setting in a natural experiment which exploits the variation created by the enactment of a federal law which extended the existing laws in selected states to all the US states. I find that the reduction in this cost

1. Lenders rejected 29.87 million applications, about 20% of all applications between 2000 and 2008. 8.34 million applications were rejected due to credit history. Lenders provided reason for about 70% of the rejected applications. Hence, the fraction of denied mortgages due to credit history is 39.4% if we only consider rejections for which a reason was provided. Debt to income ratio, the second most frequent reason for rejection, affects rejection of just half as many applications as does credit history.

2. SCF 1998, 2001, 2004 and 2007.

3. In a frictionless credit market with perfectly informed consumers (and lenders), (1) creditunworthy consumers *would not apply* for credit because it is costly to apply, and it is costlier when the application is rejected; (2) no creditworthy consumer, who needs credit, would choose to *not apply* for credit.

results in increase in consumer demand for mortgage by ~$27.7 billion, and increase in successful origination by ~$2.4 billion. The increased demand was for the owner-occupied housing, and came from seemingly good quality consumers – these mortgages were extended in *ex-ante* more creditworthy areas, prime consumers, and were less likely to be defaulted upon even during and after the financial crisis of 2008. Furthermore, the approval ratio increase is primarily observed among populations with higher education, and in the lower income quartile. The increased demand due to free credit reports does not result in statistically significant increase in house prices, but does increase the growth rate of house prices by ~2.4% in the treated areas with marginal statistical significance.

I interpret these results through a channel of increased *self-learning* among the consumers. Three findings support this channel: (a) mortgages become less likely to be rejected due to credit history; (b) consumers' tendency to credit-shop increases; and (c) the fraction of first-time homebuyers in the market increases by 1%. All in all, these findings highlight the hitherto understudied role of the economic cost of obtaining one's own credit report in shaping actions of consumer in the credit markets.

Lower cost of obtaining one's own credit report can impact the demand for credit and the approval ratio through multiple channels. First, consumers can use the information contained in the credit report prior to a credit application to better self-assess their creditworthiness. The report can aid in better self-assessment because it usually contains information on the consumer's credit history, and also, some indication of the available borrowing capacity of the consumer, as shown in a sample credit report in Figure (1). Low financial literacy and complexity of debt products act as barriers to accurate self-assessment of creditworthiness. Brown, Haughwout, Lee, and Van Der Klaauw (2011) shows that consumers underestimate their existing debt, and Perry (2008) estimates that as many as 32% of consumers overestimate their credit ratings, while only 4% underestimate it.[4] Self-assessment of creditworthiness matters because incorrect self-assessment leads to worse financial outcomes (Courchane, Gai-

4. Consumers underestimate their student (credit card) debt by as much as 25% (37%) (Brown et al., 2011).

ley, & Zorn, 2008). Second, accessing the report before applying for credit provides the consumer an opportunity to review the decision, especially if the report contains credit-relevant negative information.[5] Third, the credit report itself may contain incorrect information. Avery, Calem, and Canner (2004) finds that in a sample of credit reports representative of the US population, about 46% of consumers had missing credit limits . For these reasons, experts, including the Federal Reserve Board, actively recommend consumers to check their own credit reports.[6]

I causally estimate the effect of reduced cost of credit history information on credit demand and matching in the credit market using changes in the regulation governing the cost of credit reports. The federal Fair and Accurate Credit Transactions Act (FACTA) of 2003 grants consumers the right to obtain a *free* credit report from each of the nationwide consumer reporting agencies once in every twelve-month period. However, seven US states — Colorado, Georgia, Maine, Maryland, Massachusetts, New Jersey, and Vermont (henceforth, the pre-FACTA states) — had already enacted state laws prior to 2004 allowing their residents to access free credit reports (Figure 2).[7] The laws are described in detail in Section (2), Institutional Background.

I utilize a DID setting in which the pre-FACTA states constitute the control group and the states surrounding the early-adopting states constitute the treatment group (Figure 3). I restrict the focus only to the counties lying along the border of the control and treatment states, thereby controlling for local socioeconomic conditions (Figure 4). Moreover, this natural experiment does not rely on law enactment by individual states, but utilizes the federal extension of the existing state laws. This empirical strategy mitigates the criticism that states' selection into treatment is endogenous.

I first estimate the effect of free credit reports on the number of mortgage applica-

5. The applicant can delay the application, take steps to improve the credit record, correct any inaccurate information therein, or altogether decide to not apply for credit.

6. "It can be especially helpful to see a copy of your credit report before you apply for, say a car loan, a mortgage, or a credit card. Errors in credit reports are not uncommon." Federal Reserve Bank of Philadelphia. https://www.philadelphiafed.org/consumer-resources/publications/what-your-credit-report-says

7. Colorado in 1997, Georgia in 1996, Maine in 2003, Maryland in 1992, Massachusetts in 1995, New Jersey in 1996, and Vermont in 1992. State residency for free credit report is determined by the same criteria as those for the tax residency for the state.

tions. Mortgage applications in a treated census tract increase by 13.2% – 15.1%. This corresponds to $4.83 million increase in mortgage demand per treated census tract, or $27.7 billion in the entire treated area. Also, the approval ratio increases by 1 percentage point in the treated census tracts relative to the control census tracts. This increase is equivalent to ~2.7 more mortgages per census tract getting approved, amounting to ~$2.4 billion increase in successful mortgage origination. These results are robust to the inclusion of $Census\ Tract$ fixed effects and "$Border \times Year$" fixed effects, and a host of time-varying controls capturing the economic conditions at county and state level. Did the demand increase because lenders started relaxing lending criteria, or did the pool of applicants in the credit market improve? In the absence of a lender-driven increases in credit supply, which I show rigorously in the last section, the mortgage approval ratio captures the change in the applicant pool.[8]

I next examine whether the increase in mortgage demand is from consumers who intend to utilize the property for occupancy or from consumers who intend to buy properties for non-occupancy purposes, such as investment in order to profit from the rising house prices. To estimate the demand increase in the owner-occupied housing category, I regress the number of applications and the mortgage approval ratio using the DID specification for the subset of mortgage applications which are for owner-occupied property only. The estimated increase in the demand for this category is 10.7% – 11.5%, which is approximately the same as that estimated for all the categories combined. I further test for any change in the mortgages in the owner-occupied category as a fraction of total mortgages and as a fraction of successful mortgages. I do not find any evidence of change in these fractions. All in all, these results suggest that the demand increase due to free credit reports was for properties intended for occupancy purposes.

I rationalize these results through a channel in which consumers *self-learn* about their true creditworthiness through free credit reports (*t*he self learning channel). Demand suppressing credit worthy consumers enter the credit market, and

8. Mortgage approval ratio is the ratio of mortgages approved or originated to total mortgages applied for in a census tract.

credit-unworthy consumers go out of the market, thereby increasing the demand for mortgage and approval ratio. I conduct three tests to examine this channel. First, I examine for any change in reason for rejection of mortgage applications. If applicants are more likely to access their credit report and take action based on that information prior to the mortgage application, then in aggregate, a mortgage should become less likely to be rejected due to credit history. I find suggestive, albeit weak, evidence of this. The fraction of applications denied for credit history declines in the treated areas, but there is no significant change in the fraction of applications denied for debt-to-income ratio.

The second test of the self-learning channel is motivated by the idea that a better-informed applicant is more likely to shop for a mortgage deal before application than a less-informed applicant (credit shopping). Consequently, the former is less likely to withdraw an in-process mortgage application than the latter. The fraction of total applications which are withdrawn while-in-process decreases by 0.9 percentage points (~ 2.48 applications per 1000 adult), saving ~US $26.5 million in the upfront application fees for the applicants, at an average cost of USD 400 per application.

In the final test of the self-learning channel, I examine the proportion of first-time borrowers in the mortgage market. As we discussed earlier, many consumers do not apply for a mortgage because they anticipate a rejection. If the fraction of consumers who are more creditworthy than they expected is more than the fraction of consumers who are less creditworthy than they expected, then we would observe an increase in the proportion of first-time borrowers in the mortgage market. Using a subset of mortgages purchased by Fannie Mae and Freddie Mac (Government Sponsored Entities or GSEs), I find that the fraction of first-time homebuyers increases in the treated areas by 1 percentage point. This suggests that, in aggregate, we have a higher fraction of consumers underestimating their own creditworthiness than overestimating it. Reducing the economic cost of credit reports leads to such consumers discovering their true creditworthiness and entering the credit market.

Next, I focus on the heterogeneity in the effect of free credit reports across the characteristics of the consumers. If the effect of free credit reports works primarily through the consumers, then the effect should vary across different groups of consumers sorted on relevant characteristics. I examine the heterogeneity across three key consumer characteristics – creditworthiness, education, and income.

First, I examine the heterogeneous effect of free credit reports across creditworthiness of the consumers. In the absence of individual-level credit score data, I turn to a location-based proxy for creditworthiness – fraction of the population in a county which is prime. Since free credit reports allow consumers to assess their creditworthiness more accurately, and lenders are more likely to approve the application of a creditworthy consumer, approval ratio should increase in the areas where higher fraction of population is prime. Taking the mean fraction of population which is prime in a region as a benchmark in 1999, following Mian and Sufi (2009), I find that the approval ratio improves in the prime counties, but not in the subprime counties.

Noting that the county-based measure of creditworthiness is noisy, I use a more precise proxy of creditworthiness available at the census tract level – the number of payday lenders. The idea is that payday lenders tend to operate in areas with a high subprime population (Prager, 2009). Using this proxy, I again find that the mortgage approval ratio increases only in the areas having high creditworthiness.

Second, I examine the heterogeneous effect of free credit reports across education. The idea behind testing for the differential effect of free credit reports across different education levels is that, *ceteris paribus*, a more-educated consumer is more likely to utilize a credit report while making a credit decision than a less-educated consumer. Using the fraction of the adult population who are graduate or equivalent degree holders as a proxy for education, I find that mortgage applications as well as the mortgage approval ratio increase in high education areas, but not in low education areas.

Last, I examine the effect of free credit reports across consumers' income. The motivation behind this test is that the economic cost of mortgage rejection is higher for the lower-income consumers. I find that the mortgage approval ratio increases by 1 percentage point in the bottom income quartile. This suggests that the creditworthy applicants in the bottom income quartile are more likely to abstain from participating in the mortgage market when the cost of learning about their own credit history is high.

Overall, these tests reveal that the effect of free credit reports is heterogeneous across the three consumer characteristics in the expected direction. These results point to a consumer-driven demand effect (demand-pull), rather than a lender-driven effect (supply-push). I explore the role of supply-push effect later.

Are these new mortgages, supposedly induced by free credit reports, more likely to be defaulted upon? For this, I compare the default rates on mortgages originated in the year of the event to those originated in the year before the event in the treated area relative to the control area (*adjusted default rate*) over six years after the origination. Consistent with the idea that free credit reports induce creditworthy consumers in the market, I find that these mortgages are less likely to be defaulted upon, even during and after the financial crisis (2007 – 2010). This result is noteworthy. Free credit reports not only result in more consumers applying for mortgages and lenders being more likely to approve them, but also the performance of these mortgages is better, *not worse*, than for comparable mortgages, even in and after the financial crisis.

I devote the last set of tests to examine if the increased mortgage origination is primarily due to an increase in demand from consumers, and not solely due to lenders increasing the supply of mortgages in response to the experiment. Specifically, lenders in the treated areas are exposed to the knowledge that their consumers are better informed about their creditworthiness. Thus, lenders may increase the supply of mortgages even if there is no real change in the pool of consumers. Admittedly, it is difficult to quantify by how much the demand would have increased if there were no changes in applicant pool, but simple lenders were willing to lend more on the knowledge that consumers are better informed. Still, we can rule out the explanation that *all* the changes observed is *solely* due to lenders simply increasing the supply of mortgages. I begin with the observation that though lenders optimally decide the mortgage approval ratio, they do not choose the number of mortgage applications, which increases in the treated areas. Further, previous results establish that the effect of free credit reports is heterogeneous in the considered characteristics of consumers – creditworthiness, income and education. To the extent that these same characteristics do not induce increase in supply of mortgages, these results point to a demand-pull effect. In order to further examine the role of lenders in increased supply of mortgages, I employ four tests.

First, I appeal to a simple supply-demand framework. Increase in the quantity of a good demanded, together with an increase in the price of that good is possible only if the demand curve shifts outwards, while the supply curve stays the same, or suffers an inadequate shift so that the price need to rise in equilibrium for the credit supply to match the increased demand. I test whether the price of mortgages increased after the experiment. Using a sub-sample of mortgages for which the interest rate data are available, I find that the mean and median interest rates increased in the treated areas relative to the control areas. This is consistent with an increase in consumer-driven demand-pull effect, and contradicts a *sole* lender-driven supply-push effect. Further, noting the fact that the supply of mortgage in the US is highly elastic due to the GSEs being a significant buyer of mortgages in the secondary market, this result underscores the demand stimulating effect of free credit reports.

Second, I utilize the heterogeneity in lender density. This is based on the idea that if the increase in mortgages were driven by lenders, we would expect higher mortgage origination and mortgage approval ratio in areas having a high density of mortgage lenders. DID coefficients for the amount of mortgage origination per adult and approval ratio in areas having a high density of lenders are statistically the same as those in areas having a low density of lenders. Thus, this result rules out the explanation that increased approvals are just a manifestation of lenders being motivated to originate more mortgages.

Third, I examine the role of securitization in increased approval ratio. It may be argued that lenders may be originating more mortgages in a bid to earn higher commissions from private securitization. If this were the true reason behind the observed increase in approval ratio in the treated areas, a higher fraction of mortgages would be sold to non-government entities. The regression results reveal that there is no change in the fraction of mortgages that are approved and sold to non-government entities, addressing the concern that private securitization might be the reason for observed increase in lending.

Fourth, I test the subprime supply hypothesis that the increase in the mortgage approval ratio is due to lenders increasing supply in subprime areas (Mian & Sufi, 2009). Previously, I showed that the effect of free credit report is pronounced in prime counties and among the prime population. In a further test, I use the subset of mortgages for which the application-level credit score is available from the GSEs. I define an applicant as prime if their credit score is more than 620. This application-level test confirms that the mortgage origination to prime borrowers increases in the treated areas, and thereby rejects the subprime supply hypothesis.

All in all, the four tests examining the role of lenders' characteristics and incentives support the fact that the increased mortgage demand and the mortgage approval ratio in the treated areas are not solely driven by lenders.

Related Literature

This paper contributes to the multiple strands of the literature. The most relevant is the nascent literature on information provision and its effect on market participants. This is the first paper to document that matching in the credit market can be improved if the cost of information about consumers' own credit history is reduced. In a field experiment, Homonoff, O'Brien, and Sussman (2019) finds that borrowers who are randomly provided information about their own FICO® scores are less likely to default. Kulkarni, Truffa, and Iberti (2018) shows that standardized financial contracts reduce consumer delinquency by 40% and sophisticated (unsophisticated) borrowers are helped the most by increased product disclosure (product standardization). Liberman, Neilson, Opazo, and Zimmerman (2018) documents aggregate welfare loss and a reduction (increase) in the cost of credit for poorer defaulters (non-defaulters) when consumer reporting agency (CRAs) are barred from reporting consumer defaults to lenders. Dobbie, Goldsmith-Pinkham, Mahoney, and Song (2016) shows that removal of bankruptcy flags from credit reports results in a significant increase in credit card balance and mortgage borrowing. In the investment market, consumers become more sensitive to expense ratios and short-term performance after regulation makes the fee and performance disclosure of 401(k) plans mandatory (Kronlund, Pool, Sialm, & Stefanescu, 2019).

This paper also contributes to the literature on financial literacy and household behavior. Low financial literacy has been shown to have detrimental economic outcomes: high mortgage delinquency and home foreclosure (Gerardi, Goette, & Meier, 2010); poor mortgage choice (Moore, 2003) and large debt accumulation (Lusardi & Tufano, 2009; Stango & Zinman, 2009). Perry (2008) reports that about one-third of the population overestimates its own credit rating, while Courchane et al. (2008) shows that inaccurate assessment leads to higher financing charges and higher probability of denial. Using a field experiment, Balakina, Balasubramaniam, Dimri, and Sane (2020) show that an educational intervention reduces the likelihood of borrowers purchasing a sub-optimal financial product. In another field experiment, Hundtofte (2017) shows that even though distressed borrowers increase repayments in response to loan modification programs, they ultimately fail to realize the financial benefit of the loan modification, due to imperfect financial sophistication and misvaluation of the contract. In this paper, I document that free credit reports result in more consumers applying for mortgages, and a higher probability of mortgage approval.

This paper further relates to the extensive literature on the mortgage expansion in the US prior to the great recession. Literature has advanced several supply side arguments for excessive mortgage expansion. Mian and Sufi (2009) argues that subprime borrowers had disproportionately large credit growth but lower income growth, and eventually suffered large mortgage delinquencies. Adelino, Schoar, and Severino (2016); Foote, Loewenstein, and Willen (2016) and Conklin, Frame, Gerardi, and Liu (2018) document contradictory results. Keys, Mukherjee, Seru, and Vig (2010) shows that the originate-to-distribute model of securitization resulted in lenders relaxing the borrower screening. In this paper, I document the increase in mortgage applications and the mortgage approval ratio due to free credit reports, which are suggestive of the demand-side channel of self-learning.

Finally, this research also speaks to the nascent literature on issues related to the information contained in credit reports. A few reports by government agencies – the Federal Reserve Board (FED) and Congressional Research Service (CRS) – and consumer advocacy groups provide a perspective on issues related to credit reports: What information is contained in credit reports (Avery, Calem, Canner, & Bostic, 2003); limitation of the credit report data and its consequences on credit (Avery et al., 2004); effect of free credit reports on the CRA industry (Nott & Welborn, 2003); and the extent of errors in credit reports and consumer loss (Cassady & Mierzwinski, 2004; Consumer Federation of America, 2002; Golinger & Mierzwinski, 1998). In this paper, one of the motivations for the increase in demand for mortgage credit is the reduction in demand suppression by consumers incorrectly anticipating rejection.

The rest of the paper is organized as follows. Section 2 discusses the institutional background. Section 3 describes the empirical setting. Section 4 provides description of the data used in this paper. Section 5 shows the results of the paper. Section 6 discusses the role of lenders in the increased demand of mortgage and mortgage approval ratio and provides evidence against it. Section 7 shows the robustness tests for the results, and Section 8 concludes the paper.

2 Institutional Background

Laws Governing Consumers' Access to Credit Reports in the US: Before FACTA, the Fair Credit Reporting Act (FCRA) governed consumer credit information-related laws in the US. Even under the FCRA, consumers had the right to see the contents of their credit report except for the credit score (Avery et al., 2003). A 1992 amendment to the FCRA mandated that the cost of disclosure of credit information should be reasonable, while a 1996 amendment to FCRA set the maximum cost at $8. Moreover, under the FCRA consumers could receive a credit report without charge under specific circumstances, for example, a consumer making a request within 60 days after receiving a notice of an *adverse action* taken against him or her on the basis of the information in the credit report.[9][10]

Even though the FCRA allowed free credit reports at the federal level under specific circumstances, it was extremely uncommon for consumers to proactively request their own credit report. Out of approximately 1 billion credit reports generated annually, only 1.6% were disclosed to consumers (Avery et al., 2004). Of these 1.6% consumer-disclosed reports, only 5.25% were proactively requested by consumers, while 94.75% were disclosed to consumers under the FCRA provisions mentioned earlier (Nott & Welborn, 2003).[11] Thus, only 0.084% of all credit reports generated were disclosed to consumers as a result of their own request.

In addition to the federal provisions under the FCRA, residents of seven states – Colorado (CO), Georgia (GA), Maine (ME), Maryland (MD), Massachusetts (MA), New Jersey (NJ), and Vermont (VT) – could access free credit reports under respective state laws enacted over the years (see Footnote 7 for the year of individual enactments). I call these seven states the pre-FACTA states. Consider, for example, the state of Colorado. It enacted a law for providing free credit reports on April 21, 1997 through The State of Colorado SENATE BILL 133. Section 4, paragraph (E) of this bill added the following to Title 12, Article 14.3-104 of the Colorado Statute:

> (*E*): *Each consumer reporting agency shall, upon request of a consumer, provide the consumer with one disclosure copy of his or her file per year at no charge whether or not the consumer has made the request in response to the notification required in*

9. Adverse action notice can be sent to a consumer by the *user* of consumer report (e.g. banks, financial institutions, insurance firms), or, a debt collection agency affiliated with the CRA stating that the consumer's credit rating may be or has been adversely affected.

10. Furthermore, consumers can receive credit report free of charge once in 12 months if he or she makes a request to the CRA for the credit report and certifies that: (A) She/he is unemployed and intends to apply for employment in the 60 day period beginning on the date on which the certification is made; (B) She/he is a recipient of public welfare assistance; (C) She/he has reason to believe that the file on the consumer at the agency contains inaccurate information due to fraud.

11. Break down of the 94.75% credit reports disclosed under FCRA provisions: 84% due to *adverse action*; 11.5% due to fraud claim; 0.4% due to unemployment, 0.1% due to consumer being on public assistance.

paragraph (a) of this subsection.

The provisions of the FCRA were to expire in 2003, leading to the enactment of the Fair and Accurate Credit Transactions Act (FACTA) on December 4, 2003. One key provision of FACTA was to provide free annual disclosure of credit reports to consumers by each of the three national credit reporting agencies. As a federal law, FACTA is applicable to US consumers from all states. Thus, residents of the pre-FACTA states can enjoy free credit reports under FACTA as well as under their respective state laws, while residents of other states can enjoy free credit reports only under FACTA.

3 Empirical Setting

The pre-FACTA states (CO, GA, ME, MD, MA, NJ and VT) all had free credit reports available for their residents prior to 2005 (see Footnote 7 and Figure 2); while all US states obtained access to free credit reports after the enactment of FACTA in 2004, and subsequent establishment of www.annualcreditreport.com in 2005.[12] To evaluate the effect of free credit reports, I designate the seven pre-FACTA states as a control group, and the states bordering these pre-FACTA states as the treatment group in a DID setting. Figure 3 illustrates the control and the treatment states on the US map.

To isolate the treatment effect of free credit reports from the effect of local economic conditions, I restrict the focus over a narrow geographic area around the borders of the pre-FACTA states. I do this by focusing only on the counties lying at the border of the pre-FACTA control states and surrounding treatment states, and removing the inner counties of the treatment and control states (*border county strategy*). Figure 4 illustrates this strategy. For example, Garrett County, Maryland (a border county from a control

12. www.annualcreditreport.com was rolled-out in four phases from Dec 2004 to Jan 2005:
Dec 1, 2004: Alaska, Arizona, California, Colorado, Hawaii, Idaho, Montana, Nevada, New Mexico, Oregon, Utah, Washington, and Wyoming.
Jan 3, 2005: Illinois, Indiana, Iowa, Kansas, Michigan, Minnesota, Missouri, Nebraska, North Dakota, Ohio, South Dakota, and Wisconsin.
Jan 6, 2005: Alabama, Arkansas, Florida, Georgia, Kentucky, Louisiana, Mississippi, Oklahoma, South Carolina, Tennessee, and Texas.
Jan 9, 2005: Connecticut, Delaware, DC, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, North Carolina, Pennsylvania, Rhode Island, Vermont, Virginia, and West Virginia.

state) and Somerset County, Pennsylvania (a border county from a treatment state) will be included in the sample, while inner counties of Maryland (e.g., Howard County) and Pennsylvania (e.g., Montgomery County) will be excluded.

The main regression specification is the following:

$$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it} \tag{1}$$

where y_{ijct} is the outcome variable for a census tract level i from a county c lying at the border of control state j in year t. Recall that there are seven control states; hence j ranges from one to seven. Key outcome variables of interest are the number of mortgage applications aggregated over a census tract per 1000 adults and the mortgage approval ratio in a census tract. $Post_T$ takes value 1 for year $t \geq 2005$.[13] $Treatment_{ic}$ is 0 for all the census tracts i in counties c which are at the border of pre-FACTA (control) states, and is 1 for all the census tracts i in counties c at the border of the treatment states. α_i controls for time-invariant and census tract-specific fixed effects. $Economic\_controls$ are county- and state- level annual variables capturing the fluctuations in local economic conditions. Variables included in $Economic\_controls$ are the annual growth rate of income per capita in the county, aggregate employment in the county, and the annual growth rate of the gross domestic product (GDP) of the state. $\gamma_{j,t}$ is the "$Border \times Year$" fixed effect. It groups together all the census tracts i which lie around the border of the same control state j, and ensures that the census tracts in a control county of a pre-FACTA state, say Colorado (state j), are compared only with the census tracts from the treatment counties *surrounding* Colorado. This means that a control census tract from Colorado does not get compared with a treatment census tract from states surrounding Georgia.[14] Further, it controls for any time-varying regional economic shock affecting

13. Though the act was passed in Dec 2003, the centralized source for free credit report was first rolled out on Dec 1, 2004. Since all four phases were rolled out from Dec 1, 2004 to 9 Jan 2005, I use 2005 as the event year.

14. The sample consists of seven control states and 20 adjacent states. To account for local time-varying economic shock, I employ the "$Border \times Year$" fixed effect. Here, a border is identified in terms of the control state. For example, the border fixed effect identifier for the control state Colorado, will take the same value for all the states sharing a border with Colorado, namely - WY, UT, AZ, NM, OK, KS, and NE.

neighboring states. I cluster the standard errors at the county level in all specifications to account for any potential correlation in error terms from census tracts belonging to the same county, and also to account for any serial correlation in the dependent variable.

This empirical strategy critically relies on the fact that residents in the seven pre-FACTA states accessed free credit reports significantly more than the bordering treatment states prior to 2005. Data on free credit report usage prior to 2004 confirms this assertion. In the Senate hearing on the FCRA, Senator Bennett reported that the use of free credit reports, relative to the national average, was 250% higher in GA, 204% higher in MD, 153% (458%)[15] higher in CO, 35% higher in NJ, and 25% higher in MA.[16] Moreover, some evidence suggests that the pre-FACTA states had better credit environments than the states with no free credit reports. Vermont, in 2002, had the lowest rate of consumer bankruptcies in the US; Massachusetts the second lowest. Similarly, the effective interest rate on a conventional mortgage in Vermont and Massachusetts are below the national median.[17]

Another advantage of this empirical setting is that it does not use the state-by-state adoption of free credit report laws, which may be an endogenous action of the state in response to prevailing local socioeconomic conditions.[18] Hence, this empirical setting mitigates the endogeneity concern that some state-specific observed or unobserved characteristics are driving the enactment of the law and the outcomes of interest.

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15. Based on free credit report usage rate data from Nott and Welborn (2003) and national average usage rate data from Senate hearings (see Footnote 16)

16. Hearings before the Committee on Banking, Housing, and Urban Affairs, United States Senate. S. HRG. 108–579.

17. Prepared statement of Joel R. Reidenberg. Hearings before the committee on Banking, Housing, and Urban Affairs, United States Senate. S. HRG. 108–579.

18. Consider the case of Vermont. In 1991, Experian (then, TRW Inc.) erroneously garbled the credit report of every citizen residing in Norwich, claiming that everyone had failed to pay property taxes. This could have resulted in banks or other lenders rejecting these residents if they applied for new credit. TRW settled the subsequent state lawsuit in 1992 and paid compensation to each affected homeowner. Vermont responded with the 1992 legislation mandating free annual credit reports for every requesting state resident.

4 Data

The key data used in this paper is the US mortgage data available under the Home Mortgage Disclosure Act of 1975 (HMDA). These data provide application-level details on applicants' demographics (race and gender), income, loan amount, type of the financial institution handling the mortgage application, outcome of the application, and geographic location of the property at the census tract-level. The time period for the study is from 2000 to 2008. I use the data until 2008 to allow for sufficient post-experiment observations, since the experiment occurs at the beginning of 2005. The sample includes mortgages for three purposes – home purchase, refinance, and home improvement.[19]

The coverage of mortgages in the US under the HMDA is the widest. It contains 190.3 million application-level observations over the sample period. I start processing this data by first removing observations for which state, county, or census tract information is missing or "NA", or state FIPS is "0", "00" or "0 ". This drops 2.5% of the observations, leaving 185.6 million mortgages with an identifiable county. Next, I drop three types of mortgages: first, I drop covered loan purchases by the financial institution from other institution (18.80%), as these mortgages are not borrower initiated transactions; second pre-approval requests denied by financial institutions (0.01%), as this data was included in HMDA reporting only from 2004;third, the pre-approval requests approved by the financial institutions but not accepted by the applicants as this data, too, has been included in HMDA only from 2004, and also as it is optionally reported (0.025%). This leaves 150.7 million mortgage applications in the sample. I aggregate these mortgages at the census tract and year level, yielding 65,976 unique census tracts and 572,512 *Census Tract* $\times$ *year* observations. Utilizing the *county adja-*

19. HMDA uses census tract definition from Census 1990 for data until 2003, and from Census 2000 for 2004 onward. To make the geographic area consistent across these two regimes, I scale the variables aggregates at the census tract level for years 2000 to 2003 from Census 1990 tract definition to Census 2000 tract definition. I use the ratio of population residing in the 1990 definition of census tract to that in 2000 definition of census tract as the scaling variable. I use the data from Census Bureau (2006) for this conversion.

cency data,[20] I select the census tracts which lie in the counties at the border of the control and treatment states (see Figure 4). These census tracts constitute the final sample of 9,821 unique census tracts – 5,724 in the treatment group and 4,097 in the control group – making up 83,236 "$Census\ Tract \times Year$"-level observations.

Though the coverage of HMDA data is the largest, it does not provide some important mortgage-level details, such as the credit score of the applicant. Hence, for tests which require these variables, I utilize data from GSEs – the Federal National Mortgage Agency (Fannie) and the Federal National Home Loan Mortgage Corporation (Freddie). The GSEs provide data on 30-year fixed rate single family mortgages purchased by them. This mortgage type is the most popular mortgage type in the US. The combined data, henceforth the GSE data, has 33 million mortgage-level observations.[21] The data consists of application-level information on debt-to-income ratio, credit score, first-time homebuyer flag, investment purpose and more. The property location is available as a 3-digit zip code (henceforth, zip3) and state. To map zip3 locations to the counties at the borders of the states, I use the 2010 Q3 version crosswalk files provided by the US Department of Housing.[22] I aggregate the mortgage-level observations to zip3-state level. This creates a panel of 225 unique zip3-states, leading to 7,711 "$Zip3 - state \times Quarter$" observations.

I use a few other data sources in this paper. For gauging the creditworthiness of a county, I use Federal Reserve Bank of New York and Equifax (n.d.) data on subprime county population. For measuring county-level economic characteristics, such as employment and number of establishments, I use data from the annual survey of County

20. US Census Bureau provides data on which counties are surrounded by which other counties. https://www.census.gov/geographies/reference-files/2010/geo/county-adjacency.html

21. The data from Fannie Mae's consist of 30-year, fixed-rate, fully documented, single-family amortizing loans that the company owned or guaranteed on or after January 1, 2000. The data from Freddie Mac consists of fully amortizing 30-year fixed-rate, Single Family mortgages, that the company acquired with origination dates from 1999 onward.

22. Areas bounded by 3-digits zip codes do not align perfectly with the county borders. Hence, to allow the regression estimates to be estimated as close as possible to the border areas when only 3-digits zip code information is available, I start with a 5-digits zip code to county crosswalk file. I remove all the 3-digits zip code combinations for which *none of the* constituent 5-digits zip codes are located within the bordering counties. URL for crosswalk data is: https://www.huduser.gov/portal/datasets/usps_crosswalk.html

Business Patterns.[23] To map the zip code-level variables from County Business Patterns to census tract-level, I use the Geocorr 2000 tool from the Missouri Census Data Center.[24] The data on state level economic conditions comes from the Bureau of Economic Analysis, and the data on population characteristics at census tract-level is from Census 2000 (Manson, Schroeder, Van Riper, & Ruggles, 2019).

Key outcome variables of interest are the number of mortgage applications; the mortgage approval ratio; the fraction of total applications which are denied for credit history or debt-to-income ratio; and the fraction of total applications which are withdrawn by the applicants while still under processing by the lender. I define an application to be successful if the mortgage has either been originated or the application has been approved but not accepted by the applicant. The mortgage approval ratio is the ratio of the number of successful applications to the number of total applications in a census tract. The other four ratios are also calculated as a fraction of total applications in a given census tract.

Table 1 provides the summary statistics of the HMDA sample. Panel A shows the summary statistics for the entire sample, while panel B shows the statistics for the pre-treatment sample. We see that the treated census tracts have fewer mortgage applications per 1000 adults, lower mortgage approval ratio, and higher fraction of applications denied due to credit history and denied due to debt-to-income ratio. The five ratios – the approval ratio, the three denial ratios, and the withdrawal ratio – do not sum to one. There are two reasons for this. First, the reporting of reason for denial is not mandatory under HMDA regulations; hence an application may be recorded as denied without any stated reason (70.81% of denied applications have at least one stated denial reason). Second, an application can have multiple reasons for denial, e.g., denied for credit history and debt-to-income ratio.

Panel B shows a comparison of the treatment and control groups in the pre-

23. County Business Patterns data is available at the URL: https://www.census.gov/programs-surveys/cbp/data.html

24. *Geocorr 2000: Geographic Correspondence Engine. Version 1.3.3 (August, 2010) with Census 2000 Geography*. http://mcdc.missouri.edu/applications/geocorr2000.html.

treatment period. We see that the differences between the treatment and control census tracts in the pre-treatment sample are similar to those observed in the full sample in Panel A. The p-value for the t-test of the difference in mean between the two groups for each variable is also shown in Panel B. Results from the t-test suggest that the control and treatment census tracts are different in pre-treatment years on these observed characteristics. This also raises the endogeneity concern that these groups may differ on unobservable characteristics. This concern can be mitigated in two ways. First, the DID setting can accommodate pre-existing differences between the treatment and control subjects. Second, the visual representation in Figure 5 seems to suggest a parallel trend: the difference in the mortgage approval ratio is not significant before the experiment, but starts to increase after the experiment. Third, in all regressions I include $Census\ Tract$ fixed effects, accounting for any time-invariant differences across the census tracts. I also include "$Border \times Year$" fixed effects, which can control flexibly for any time-varying regional shock that impacts neighboring states.

One may argue that the enactment of FACTA in Dec 2003, and subsequent establishment of www.annualcreditreport.com, is not a significant event, and thus the natural experiment is invalid. I address this concern in two ways. First, I plot the search interest for the key phrase *Free Credit Report* in Figure 6 from 2004 to 2010.[25] Search interest shows a significant peak in Jan 2005, when the website was established, suggesting that there was significant consumer interest in free credit reports in 2005.

Second, I show in Figure 7 that the popularity of the key phrase *Free Credit Report* increased significantly in the treatment states after the event, but there was no difference in popularity before the event. To do this, I begin by sourcing the *interest by subregion* data from Google Trends for the key phrase *Free Credit Report* for each year from 2004 to 2008. I calculate the mean of the popularity rank for the treatment and the control states

25. Search interest numbers are standardized index representing search interest at any time relative to the highest point during the time period of the analysis, over a given region(US in the present case). A value of 100 is the peak popularity for the term. A value of 50 means that the term is half as popular. A score of 0 means there was not enough data for this term. Since Google Trends data starts from January 2004, past data is not available.

in each year.[26] Then, I calculate $Popularity_{trt} - Popularity_{ctrl}$ for each year, the resulting plot for which is shown in Figure 7. The plot suggests that the key phrase *Free Credit Report* was equally popular in both the treatment and control states in pre-event year 2004, but in the treatment states, it became more popular after 2005, the experiment year.[27]

5 Results

In this section, I provide the empirical results of the study. I first provide the baseline results, then present the results highlighting the plausible channels. Then, I show the results highlighting the heterogeneous effect of free credit reports across different characteristics of consumers. At the end, I discuss the performance of the mortgages in the treated areas after the advent of free credit reports.

Baseline Results

The baseline results are obtained using regression Equation (1). The outcome variables of interest are the number of applications per 1000 adult population and the mortgage approval ratio, each calculated at census tract-level. All specifications use *Census Tract* fixed effects and "$Border \times Year$" fixed effects. As discussed earlier, *Census Tract* fixed effects control for any pre-existing difference across different census tracts. "$Border \times Year$" fixed effects serves two purposes. First, they control for any shock that might affect different regions of the US at different times, e.g., a shock affecting Colorado and surrounding states in the year 2003, but affecting Georgia and surrounding states in 2005. Second, it ensures that the census tracts in a control county of a pre-FACTA state,

26. Google calculates the *Popularity* of a region on a scale from 0 to 100. Value of 100 represents the location with the highest popularity of the search term as a fraction of total searches in that location. A value of 50 indicates a location which is half as popular. A value of 0 indicates a location where there was not enough data for the search term.

27. It is important to note that the methodology Google uses to calculate the popularity of a keyword in a region implies that increase in popularity of one area would mechanically reduce the popularity in other areas. Also, we cannot interpret how much the popularity increased over the year, as the popularity rank is re-based to 100 and is assigned to the region where the keyword is most popular, every year.

say Colorado, are compared only with census tracts from treatment counties *surrounding* Colorado, and it does not get compared with treatment counties surrounding any other pre-FACTA states, say Georgia.

Table 2 shows the results for baseline regression. Column (1) and (2) shows the regression of total applications per 1000 adult population as the dependent variable. The coefficient of interest is "$Treat \times Post$". It captures the change in number of mortgage applications in treatment counties relative to control counties after the free credit report becomes available in the treatment counties. In column 2, economic controls (county-level annual growth rates of income per capita and aggregate employment, and state-level growth rate of GDP) are added. We see that a treatment census tract sees an increase of 11.61 – 13.52 mortgage applications per 1000 adult. This is 11.6% – 13.5% increase in mortgage applications (over the pre-treatment average of 100.56 in treated census tracts). Average mortgage size in the treatment counties in the pre-treatment period is ~$155,500. Thus, the consumer demand for mortgages increased by $1.8 million per 1000 adults, by $4.83 million per treated census tract, or, by $27.7 billion in the entire treated area after free credit report became available.

Coefficients "$Treat \times Post$" in column (3) and (4) estimate the effect of free credit reports on the change in the mortgage approval ratio. The mortgage approval ratio increases by 1% in the treatment counties after free credit reports become available. This increase represents ~2.7 more successful applications per treated tract, 15,454 more successful applications in the entire treated area, or a ~$2.4 billion increase in successful mortgage origination across all treated census tracts. In the absence of a supply-side effect, which I confirm in Section 6, we can interpret the increase in the mortgage approval ratio as the improvement in the borrower pool.

Next, I examine whether the observed increase in the mortgage demand and the mortgage approval ratio is utilized to finance properties for occupancy purposes, as opposed to investment purposes. For this, I re-estimate the baseline regression for only the mortgages in the owner-occupied category. Panel A of Table (3) shows the results. The estimates are broadly similar to the baseline specification. Mortgage applications increase by 10 – 12% and the mortgage approval ratio increases by 1 percentage point in the treated areas relative to the control areas. Further, I examine whether the composition of applications and approved mortgages changes between owner-occupied and non-owner-occupied categories. For this, I regress the fraction of total mortgages which are non-owner-occupied, and the fraction of successful mortgages which are non-owner occupied. Panel B of Table (3) shows the results. Column (1) and (2) shows that there is no change in the fraction of mortgage applications that are not for occupancy purposes. Column (3) and (4) shows that there is very weak evidence of an increase in the fraction of non-owner-occupied mortgages by 1 percentage point for successful applications. Nonetheless, about 86% of the mortgages in the HMDA sample are for occupancy. Hence, the 1 percentage point increase in the fraction of mortgages which are in non-occupancy category is economically small. Thus, we can conclude that free credit reports led to an increase in mostly occupancy-related demand.

Channel

I now examine three mechanisms of *self learning* through which free credit reports can affect the mortgage demand and the mortgage approval ratio – credit history improvement, credit shopping, and new borrowers.

Credit History Improvement

If applicants are more likely to access their credit report after FACTA, then applicants are more likely to take remedial actions (if any) before applying for a mortgage. Thus, in aggregate, mortgage applications by *credit history-informed* borrowers is less likely to be denied for *credit history* reasons, than for other reasons. Hence, we should see a drop in the fraction of mortgage rejection due to credit history, and we should not see any significant change in fraction of non-credit history related mortgage rejections, say debt-to-income ratio.[28] I test these predictions next.

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28. Of the 150.7 million mortgages in HMDA over 2000–2008, 29.87 million (19.82%) were denied. Out of the total denials, 21.15 million (70.81%) mention the reason. Of the mortgages which mention the denial reason, the two most frequently stated are credit history (8.34 million, 39.43%) and debt-to-income ratio (4.05 million, 19.15%).

I regress the fraction of total applications denied due to credit history and debt-to-income ratio using Equation (1). I estimate this specification for all census tracts, and for the subgroup of census tracts for which denial rates (per 1000) are higher in the pre-event year 2004 than the *regional mean*.[29]

Table 4 shows the results. We see that the fraction of mortgage applications denied due to credit history decreases in the treatment census tracts relative to the control census tracts, but it is significant only in areas which had high denial rates prior to the experiment. A plausible reason for the treatment effect to be significant only in the high denial areas is that we cannot estimate the reduction in denial due to a certain reason if mortgages are not likely to be denied in the first place. We also see that there is no significant change in the likelihood of a mortgage being denied due to debt-to-income ratio. These results, albeit weak, are suggestive of the credit history improvement mechanism.

A limitation of this test is that, under HMDA regulations, reporting the reason for denial is not mandatory. This limitation is not a concern for two reasons. First, even though reporting the denial reason is optional, the compliance rate is 70.81% over 2000–2008 (see Footnote 28). Second, these results are unbiased to the extent to which it is unlikely that the incentive to report denials is not changing for HMDA lenders across the treatment and the control census tracts in 2005, the experiment year. Thus, free credit reports seems to suggest that consumers could take steps in improving their credit history.

Credit Shopping

A consumer has a right to withdraw an ongoing mortgage application. If she withdraws it before the lender has made the credit decision on the application, the application is recorded as *withdrawn* in the HMDA dataset. Such withdrawals are costly for

29. For clarity on calculation of regional mean, consider the control state Colorado (CO) and all the surrounding treatment states. Regional mean for the denial rate is the average of the denial rate across census tracts in all the counties at the border between CO and WY, UT, AZ, NM, OK, KS and NE. A census tract is then classified as a "higher denial area" if its denial rate is more than the regional mean. These steps are repeated for all seven control states, thereby dividing the entire sample into higher and lower denial areas.

the applicants, as they result in forfeiture of the upfront fees, which can be as high as US$ 400.[30] Yet, on average, 12% of the total mortgage applications are withdrawn while lenders are still processing it and have not made the credit decision.

Anecdotal evidence suggests that the most common reason for such withdrawals is credit shopping – a consumer withdraws an ongoing mortgage application because she has secured a mortgage offer at better terms from another lender.[31] An applicant armed with knowledge of her own credit history is more likely to do credit shopping *before* making the mortgage application than a consumer with no (less) knowledge of her own credit history. Consequently, the former is less likely to withdraw an ongoing application than the latter. If this is true, then in aggregate, the fraction of total applications which are withdrawn while in-process should decrease.

Table 5 shows the results of regressing the fraction of total mortgages which are withdrawn while in-process using Equation (1). We see that the fraction decreases by 0.9 percentage points in the treatment counties relative to the control counties. This represents ~2.48 fewer in-process withdrawn applications in a census tract, or 66,174 less withdrawn applications over all treatment counties, saving applicants US $26.5 million in upfront fees.

First-time Homebuyers

Many consumers in the credit market do not apply for credit in anticipation of a rejection. Many of such consumers may be creditworthy and may be overestimating the probability of rejection. Free credit reports can reduce the extent of this imperfection by aiding the consumers in better assessing their credit worthiness. I test this claim by observing the change in the fraction of first-time homebuyers in the mortgage market after the experiment. Information on whether an applicant is a first-time homebuyer is

30. https://www.reddit.com/r/personalfinance/comments/38k1l5/withdrawing_a_mortgage_application/

31. Applying to multiple lenders in a short period of time is not penalized by CRAs. For example, Experian, one of the three CRA encourages applicants to credit shop. "If you're shopping for a new auto or mortgage loan or a new utility provider, the multiple inquiries are generally counted as one inquiry for a given period of time. The period of time may vary depending on the credit scoring model used, but it's typically from 14 to 45 days. This allows you to check at different lenders." https://www.equifax.com/personal/education/credit/report/understanding-hard-inquiries-on-your-credit-report/

not available in the HMDA data, so I use the combined GSE data for this test. As previously explained, the GSE data is a subset of mortgages in the US consisting of 30-year fixed-rate mortgages purchased by Fannie Mae and Freddie Mac only, and is available at the 3-digit zip code level. Moreover, 6.71% of the observations in this data cannot identify if the applicant is a first-time homebuyer or a repeat buyer. Hence, I calculate the fraction of first-time homebuyers using alternate denominators: all GSE mortgages, or all GSE mortgages with non-missing information on the first-time homebuyers. The geographic aggregation for this regression is zip3-state-level (unlike previous regressions which were at census tract-level). I use the following equation:

$$y_{zcjt} = \beta_0 + \beta_1 Treament_{zc} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{ct} \tag{2}$$

Here, z indexes the areas delineated by 3-digit zip-codes and states. Other terms are the same as in Equation (1). Table 6 shows the results of regressing the fraction of first-time borrowers using Equation (2). We see that the percentage of first-time homebuyer increases by 1 percentage point in the treatment zip3-state area relative to the control zip3-state area. One may have a concern that the data used in this regression is a selected sample of the mortgages that were purchased by the GSE's only. For this to be a valid concern, GSE's incentives, or mandate, for purchasing first-time homebuyer mortgages relative to overall purchase would have to change in the treatment area, but not in the control areas, in the year 2005. This seems unlikely to be the case.

Heterogeneous Effect of Free Credit Report

Effect of Free Credit Report: Role of Creditworthiness

I argue that free credit reports aid in self-assessment of creditworthiness and improves the approval ratio in the mortgage market. If after the availability of free credit reports, consumers apply for credit if they are more confident of their own creditworthiness, in aggregate, the likelihood that a given application is creditworthy should increase. Thus, the areas where prime population fraction is high, should see credit approval ratios to rise. I test this prediction next.

To examine this, I begin with Federal Reserve Bank of New York and Equifax (n.d.) data on the percentage of population which is subprime (with credit score <660) in a county. I classify a county into high (low) creditworthiness category if its prime population fraction is higher (lower) than the *regional mean* (see Footnote 29). For this classification, I use data from the year 1999. This is because the creditworthiness of the population in a county might endogenously change with the onset of a housing boom. Mian and Sufi (2009) suggests that such classifications should be done at a time well before the start of a housing boom.[32]

Table 7 shows the results of regression in Equation (1) for the number of applications per 1000 adults and the mortgage approval ratio, estimated separately for prime and subprime counties. The estimates suggest that the number of applications and the mortgage approval ratio significantly increased in the treated prime counties relative to the control prime counties, but not in the treated subprime counties relative to the control subprime counties. These results provide two insights. First, free credit reports seem to result in an increase in the number of applications in prime areas. Second, excessive supply of credit by lenders in subprime areas is not driving these results because we do not see a statistically significant increase in such areas.

Nonetheless, the county-level measure of creditworthiness in the above test is noisy. Hence, I redo the above test using a more precise proxy for the creditworthiness of a locality – the number of establishments by payday lenders. The idea is that payday lenders tend to locate themselves in subprime areas (Prager, 2009). Also, the location of payday establishments is available at a 5-digit zip code-level from the Survey of County Business Patterns, allowing a cleaner identification of creditworthiness. However, many states restrict payday lending activities (Prager (2009) and Bhutta (2014)). Among the seven pre-FACTA states, only Colorado and the bordering states allow unrestricted payday lending activity. Thus, I conduct this test only for all the census tracts in the counties at the border between CO, the control state, and WY, UT, AZ, NM, OK, KS and NE (the treatment states). I classify these census tracts into two sub-samples

32. Mian and Sufi (2009) uses year 1996 to classify zip codes into prime and subprime. I use the year 1999, as this is the earliest year for which the data is publicly available.

using the average number of payday lenders in census tracts in bordering counties of these states in the pre-treatment year 2004.

Table 8 shows the results of estimating the regression in Equation (1) separately for the census tracts with a low and high number of payday lenders (high and low creditworthiness). The estimates suggest that even though the applications increase in both the high and low creditworthy census tracts relative to the control tracts, the mortgage approval ratio significantly increases only in the high creditworthy areas. These results corroborate the earlier finding that the mortgage approval ratio increases in the prime counties, as identified using the county-level measure, suggesting that the increased mortgage approval ratio is being enjoyed by the consumers from the prime location.

Effect of Free Credit Report: Role of Education

Now, I investigate the role of consumers' education on the heterogeneous effect of free credit reports. Debt products are complex, and thus, the information in the credit report may be complex as well. Thus, we should expect the effect of free credit report on the mortgage approval ratio to be stronger for more educated consumers than for their less educated counterparts.

To test this hypothesis, I start with calculating the fraction of adult population, aged between 18 and 64 years, having a graduate or equivalent education in each census tract using data from Census 2000. A person is classified as a *Graduate* if he or she has an associate degree, a bachelor's degree, or a graduate *or* professional degree. I calculate the *regional mean* of the fraction of graduates in a census tract in each region (see Footnote 29). A census tract is then classified as high-education if its fraction of graduate population fraction is higher than the *regional mean*.

Table 9 shows the results of regressing the number of applications per 1000 adults and the mortgage approval ratio in high- and low- education areas, estimated separately using Equation (1). We see that the number of applications as well as the mortgage approval ratio increase significantly in the high-education treatment areas relative to the high-education control areas, but it does not increase significantly in the low-education treatment areas relative to the low-education control areas. This finding is consistent with the enabling role of education in accessing credit – the better-educated population can assess the information in the credit report better and enjoys higher mortgage approval ratio than the less-educated population, leading to an improved applicant pool.

Effect of Free Credit Report: Role of Income

The costlier a mortgage rejection is for a consumer, the higher the effect of free credit reports. Here, I investigate the effect of free credit reports across income groups. For this, I divide the mortgage applications each year into four income groups (income quartiles) and calculate the number of applications per 1000 adults and the mortgage approval ratio.

Table 10 shows the results of regressing the number of applications per 1000 adults and the mortgage approval ratio for each income quartile separately using Equation (1). We see that the increase in application effect is concentrated among the upper three income quartiles, but the increase in the mortgage approval ratio is significant among the consumers in the lowest income quartile. This is consistent with the idea that the cost of rejection is higher for low-income consumers. Even though the effect of free credit reports on the approval ratio for the high-income-quartile consumers is ambiguous, overall, we can reasonably infer that the borrower pool improves in the lower quartile.

Mortgage Performance after the Free Credit Reports

So far, we have seen that free credit reports resulted in increased mortgage demand and a higher mortgage approval ratio in the treated areas relative to the control areas, and this effect was stronger in the prime areas. This suggests that these mortgages were extended to the ex-ante creditworthy population. But how did these mortgages perform ex-post? It is plausible that free credit reports lead to an increase in the fraction of *just-marginal* borrowers in the market, which in turn may result in a higher rate of mortgage defaults in the treated areas. Another possibility is that the consumers in the treated areas become more aware of their credit situation after the advent of free credit reports. If this is true, then mortgages from the treated areas should become less likely to be defaulted upon after the experiment. So I ask which of these two predictions holds in the data?

I examine the ex-post performance of the mortgages in the GSE data. I examine the default rate, defined as the fraction of total mortgages for which the scheduled payments are missed by 30–59 days for the first time in a given month after origination. I compare the mortgages originated in the post-event year 2005 ($\text{Def}_{2005,age}$) and the pre-event year 2004 ($\text{Def}_{2004,age}$) in the treated area, and I control for the time trend in the mortgage default rate using the mortgages originated in the control areas in same time-period. I define *adjusted default rate* as follows:

$$Adjusted\ Default\ Rate_{age} = (Def_{2005,age} - Def_{2004,age})_{treated} - (Def_{2005,age} - Def_{2004,age})_{control} \quad (3)$$

Here, mortgage age is measured in months since origination. The performance of the mortgage is tracked over the six years after its origination.

Figure 8 shows the plot of adjusted default rate with age. A negative adjusted default rate means that the mortgages from the treated areas are less likely to be defaulted upon than those from the control areas. The plot reveals that for most of the months with-in the six years after origination, mortgages in the treated areas are less likely to be defaulted upon than those from the control areas. Mean adjusted default rate is -0.012% over the six years, and is statistically significant at a 1% level (p-value = 0.0000). This means that, on average, the mortgage in the treated area is 0.012% less likely to be defaulted upon in any given month over six years after origination, relative to a mortgage from the control area.

Another important observation from this plot is the good performance of these mortgages after the financial crisis of 2007. A mortgage age greater than 48 months in the plot corresponds to the bust years post the financial crisis (48 months is year 2009, measured from 2005). We see that even during the bust years, the mortgages from the treated areas were less likely to be defaulted upon than those from the control areas. Overall, we observe that free credit reports induced more mortgage demand and resulted in lower default rates. Increased mortgage demand, together with a higher mortgage approval ratio and lower ex-post default rate, suggests better matching in the credit market.

Effect on House Prices

Since free credit reports lead to increase in demand for mortgages, a natural question then arises, what is the effect of free credit reports on house prices? For this, I use the publicly available house price index of Bogin, Doerner, and Larson (2016). I use the 2000 version of the index — the value of this index is set to 100 in 2000 for all the census tracts for which data is available.

Table 11 shows the results of regressing the house price index and change in house price index using Equation (1). The dependent variable in column (1) and (2) is house price index. The coefficient on "Treat $\times$ Post" is not significant. This suggests that there was no significant increase in price after the free credit reports. However, this coefficient in column (3) and (4), in which the dependent variable is change in house price index, is marginally significant. This suggest that the growth rate of house price increased in the treated areas by about 2.4% after the free credit reports.

6 Alternative Explanation: Supply-push Effect

In this section, I investigate the role of the supply-side in increased mortgage approvals and origination (supply-push effect). One alternative explanation for the increased mortgage origination is that it's not due to a consumer-driven increase (no demand-pull effect), but it is rather due to a lender-driven increase in supply of mortgages. This is plausible because the same experiment, which treated the consumers, also treated the lenders with the knowledge that their applicant pool of consumers is now better credit-informed. So, in response, lenders could adjust their lending criteria, say by increasing the approval rates. In the limit, it may be argued that the entire observed increased quantity of the mortgages is simply due to lenders increasing the mortgage supply, and not due to consumers becoming better informed and demanding more credit.

In order to tackle the *sole* supply-push explanation, I begin with the observation that the number of applications increased in the treated areas. This points to a consumer-driven demand effect. Further, the effect of free credit reports is heterogeneous in the characteristics of the consumers – creditworthiness, education and income. This suggests that the effect is varying with the characteristics of the consumers. To the extent that these same characteristics do not correlate with lender's incentive to extend more mortgages, these effects too suggest demand-pull effect. But these arguments are not sufficient.

To provide a more rigorous evidence that there was definitely an increase in consumer-driven demand, I appeal to the simple supply and demand framework. In this framework, an increase in quantity of a good demanded, together with an increase in its price, is possible only if the demand curve shifts outward, and the supply curve either stays the same, or shifts outwards or inwards inadequately such that the price of the good needs to rise in equilibrium to match the increased demand. Thus, in the context of this paper, if we find that the price of the mortgages increased, and the quantity of mortgage increased, which we have already seen, we can reasonably be sure that the demand curve shifted outwards, either with or without a shift in the supply curve.

I use the GSE data to examine the change in pricing of the mortgages, as HMDA data do not interest rates of the mortgages. I regress the mean and median interest rate in a zip3-state area using Equation 2. Table 12 shows the results. We see that mean (median) interest rate in the treated areas increased by 1.1–1.2 (1–1.1) basis points. This is consistent with an outward shift of demand curve, confirming the consumer-driven demand-pull effect as a result of the availability of free credit reports, accompanied with either no outward shift in supply curve, or an inadequate outward shift in the supply curve. We see that the increase in the mean and median interest rates is small. Still, these results are worth noting for a few reasons: firstly, the supply of mortgages in the US is highly elastic because a significant majority of these mortgages are purchased by the GSEs; secondly, there is very little variation in the pricing of these standard 30-years mortgages as compared with other credit instruments, such as credit cards or auto loans; and thirdly, these coefficients are positive and statistically significant, while even a non-significant coefficient would indicate an equal outwards shifts in the demand and the supply curve, ruling out the presence of an *sole* supply-push effect.

Second, I investigate if the effects are heterogeneous in the lenders density in a census tract. I test whether the areas having a higher density of lenders saw higher origination and approval ratios compared to the areas having a lower density. If the increase in mortgage origination were driven by lenders, we would expect more mortgages to have been originated, and likelihood of approval to have increased in areas where the density of lenders is high than in those areas where the density of lenders is low. I calculate the density of lenders as the number of HMDA lenders per adult in each census tract in the pre-event year 2004. I classify a census tract to have a high density of lenders if this density is more than the *regional mean* (see Footnote 29).

I regress the volume of mortgages originated and the mortgage approval ratio using the regression in Equation (1) separately for the areas with high and low density of lenders. Table 13 shows the results. Columns 1 through 4 show the results for total amount of mortgages originated (in 1000 USD) per adult in a census tract. We see that the coefficient of $Treat \times Post$ is smaller in specification (2) than in specification (1), and smaller in (4) than in (3). These estimates suggest that the increase in mortgage origination in the high-lender-density areas in the treated group is smaller than the increase in the low-lender-density areas in the treated group, controlling for concurrent changes in the control group. At the bottom of the table, I report the t-test for the difference in coefficient of the interaction term $Treat \times Post$ in high- and low- lender-density areas $(High - Low)$. The results confirm that the increase in mortgage origination in high-lender-density areas is not statistically different from that in low-lender-density areas. In columns 4 through 8, I repeat the analysis for the mortgage approval ratio. The results are similar – there is no statistical difference in the increase in the mortgage approval ratio in areas with a high or low lenders' density. These findings are inconsistent with the explanation that mortgage origination and mortgage approval ratio could have increased solely due to lenders increasing the supply of mortgages.

Third, I examine if the mortgage approval ratio increased due to private (non-government) securitization of mortgages. This is a plausible alternative explanation as Keys et al. (2010) shows that lenders did lax screening for the just prime borrowers (credit score≥620) and sold these mortgages to the private securitization entities prior to the financial crisis of 2007. If the argument that the reason for increased approval is private securitization holds, then we should observe a higher fraction of originated mortgages to be sold to non-government/private securitization entities in the treated areas.

To examine the above prediction, I regress the fraction of total applications which lenders originated and (1) sold to non-government entities, (2) sold to the four GSEs (Fannie Mae, Freddie Mac, Ginnie Mae, and Farmer Mac), and (3) did not sell, using the specification in Equation 1. Table 14 shows the results. The coefficients in column 1 and 2 show that there is no change in fraction of mortgage applications sold to the non-GSEs. The coefficients in column 3 and 4 show that the fraction of mortgage sold to the GSEs increased, while those in Column 5 and 6 show that the fraction of mortgages unsold did not change. These findings rule out the role of private securitization in explaining the observed increase in mortgage origination in the treated areas.

Finally, I explore the alternative argument, motivated by Mian and Sufi (2009), that lenders increased supply to subprime areas, characterized by the subprime population fraction at the zip code level. It can be argued that the increased mortgage approval ratio is due to lenders increasing credit supply to subprime borrowers in the treatment areas, which we would erroneously interpret as an improvement in the borrower pool. Results in Table (7) and (8) show that the effect of free credit reports is significant in the prime counties (census tracts) and not significant in subprime counties (census tracts).

I examine whether new mortgages in the treated areas were more likely to be extended to prime consumers than to subprime consumers using the credit scores of the applicants. Since the HMDA data do not provide applicants' credit score, I use a subset of the originated mortgages available from the GSEs, as described in detail earlier. I regress the number of GSE-purchased mortgages in the treatment and control zip3-state areas separately for prime (credit score ≥ 620) and subprime consumers. Table 15 shows the results. The coefficient in specification (1) shows that the number of mortgages extended to the prime consumers increased (by 240) in the treatment zip3-state areas relative to the control zip3-state areas. This shows that free credit reports resulted in increased mortgages for prime borrowers. It should be noted, however, that the magnitude estimated in this table is not directly comparable with those estimated in the previous tables using the HMDA data, because the observation unit is the zip3-state in the current regression, while it is the census tract in the earlier regressions. Moreover, a caution is warranted in interpreting the results in Table 15. Since this sample is a selected sample of GSE-purchased mortgages only, any increase or decrease in mortgage number and amount reflects the combined effect of any demand-side factor and changing incentives of GSEs to purchase the prime and subprime mortgages. However, it is not highly likely that the incentive of the GSEs to purchase the prime mortgage increased in the treated areas more than in the control areas in 2005.

7 Robustness

The natural experiment utilized in this paper takes place in the year 2005. The years included in the analysis are from 2000 to 2008 to allow for sufficient post-experiment observation. As the subprime mortgage origination has been argued to be the main reason for the financial crisis of 2008, it is a valid concern that the results in this paper might be due to the later years 2007 and 2008. To mitigate this concern, I re-estimate all the tests in this paper by excluding 2007 and 2008. The results are provided in the Appendix A. The re-estimated results are qualitatively and quantitatively similar to those estimated earlier for almost all specifications, despite having only two post-experiment observations (corresponding to the years 2005 and 2006). Hence, we are reasonably confident that the results documented in the paper are not driven by excessive lending prior to the financial crisis.

8 Conclusion

Credit history plays an important role in accessing financing. Does the high economic cost of obtaining one's own credit history information exclude some consumers from accessing financing? This is an important and relevant question for the US as well as for other countries which may have less-developed retail credit markets and where the processes required to obtain one's own credit report might be obscure, or may not exist at all.

In this paper, I evaluate the causal effect of reduced economic cost of credit reports on the retail credit market. I use the enactment of the US federal *Fair and Accurate Transactions Act 2003* (*FACTA*), allowing free credit reports for all consumers, as a shock to the availability of credit reports. Prior to FACTA, seven states had laws allowing its residents to obtain free credit reports. I use a difference-in-differences (DID) setting in which the border counties of the early-adopting states constitute the control group, and the border counties of the surrounding states constitute the treatment group. This empirical setting avoids the usual endogeneity criticisms of the DID settings which rely on state-specific enactment of laws.

I show that the reduced cost of credit reports results in increased mortgage applications, an improved applicant pool, and increased fraction of first-time homebuyers. The improved pool effect is observed in high creditworthiness and education areas, and among the low-income quartile population. Overall, the findings highlight that reducing the economic cost of obtaining information on their own credit history allows more consumers to access financing from the credit markets.

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Figure 1:
A Sample Credit Report

This figure shows the summary page of a credit report obtained from the website www.annualcreditreport.com for free under the Fair and Accurate Transaction Act of 2003. The specific credit history-related details are not shown. The report contains, among other things, the details of consumer's active accounts, debt-to-credit ratio, and an indication of the available credit capacity.

1. Summary

Review this summary for a quick view of key information contained in your Equifax Credit Report.

Report Date	Apr 14, 2020
Credit File Status	No fraud indicator on file
Alert Contacts	0 Records Found
Average Account Age	5 Months
Length of Credit History	8 Months
Accounts with Negative Information	0
Oldest Account	DISCOVER BANK (Opened Aug 29, 2019)
Most Recent Account	AMERICAN EXPRESS (Opened Jan 10, 2020)

Credit Accounts

Your credit report includes information about activity on your credit accounts that may affect your credit score and rating.

Account Type	Open	With Balance	Total Balance	Available	Credit Limit	Debt-to-Credit	Payment
Revolving	2	2	$606	$11,044	$11,650	5.0%	$70
Mortgage							
Installment							
Other							
Total	2	2	$606	$11,044	$11,650	5.0%	$70

Other Items

Your credit report includes your Personal Information and, if applicable, Consumer Statements, and could include other items that may affect your credit score and rating.

Consumer Statements	0 Statements Found
Personal Information	3 Items Found
Inquiries	2 Inquiries Found
Most Recent Inquiry	DISCOVER BANKAug 27, 2019
Public Records	0 Records Found
Collections	0 Collections Found

Figure 2:
States Providing Free Credit Reports prior to FACTA (pre-FACTA states)

This figure shows the seven US states which had enacted free credit reports law – Colorado, Georgia, Maine, Maryland, Massachusetts, New Jersey, and Vermont – in 1997, 1996, 2003, 1992, 1995, 1996, and 1992, respectively. These seven states are referred to as the pre-FACTA states or the control states. In December 2003, with the enactment of the Fair and Accurate Credit Transactions Act (FACTA), free credit reports became mandatory in all US states. The website www.annualcreditreport.com was subsequently established in Dec 2004 to distribute the free credit reports (see Footnote 12).

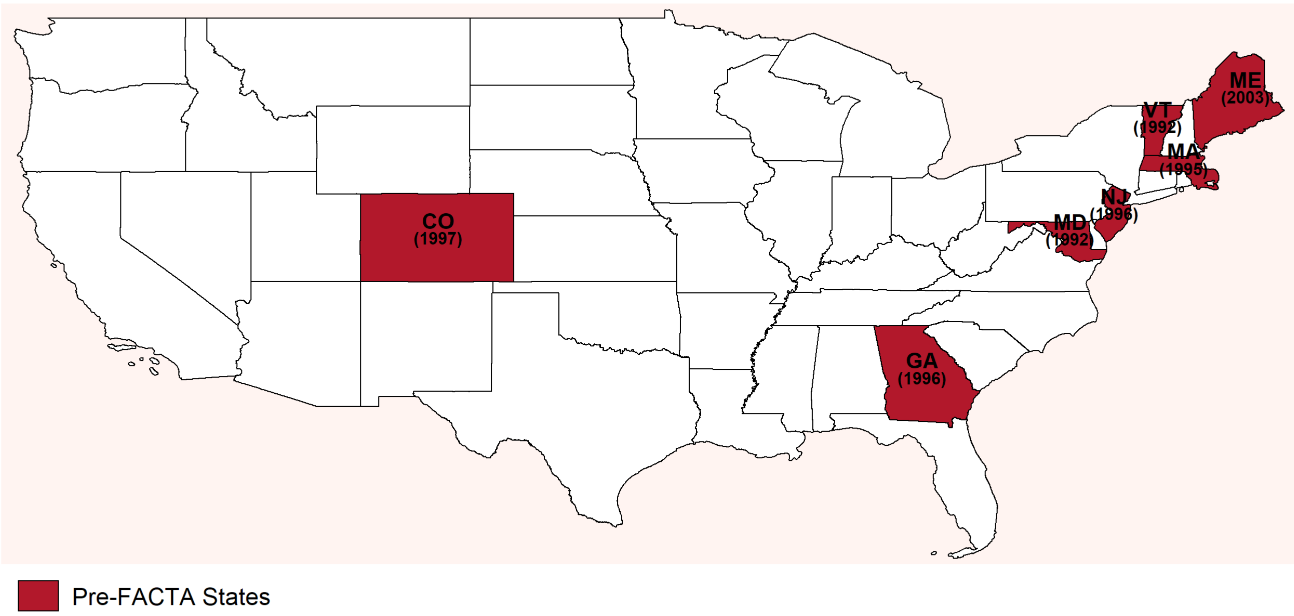

Figure 3:
Empirical Setting: Control and the Treatment States

This figure shows the seven pre-FACTA states, serving as the control states, and the 26 states that surround the control states, serving as the treatment states.

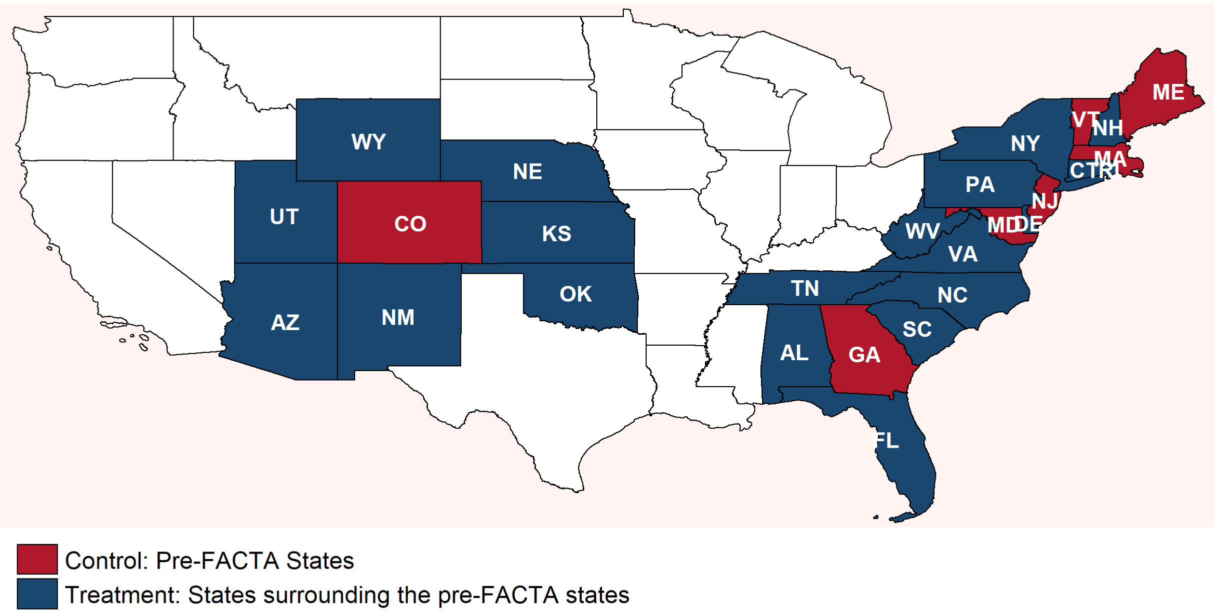

Figure 4:
Empirical Setting: Control and the Treatment Counties

This figure shows the treatment and the control counties. All the counties that lie at the border of the seven pre-FACTA states constitute the control states. All the counties from the states surrounding the pre-FACTA states and lying at the border between them constitute the treatment counties.

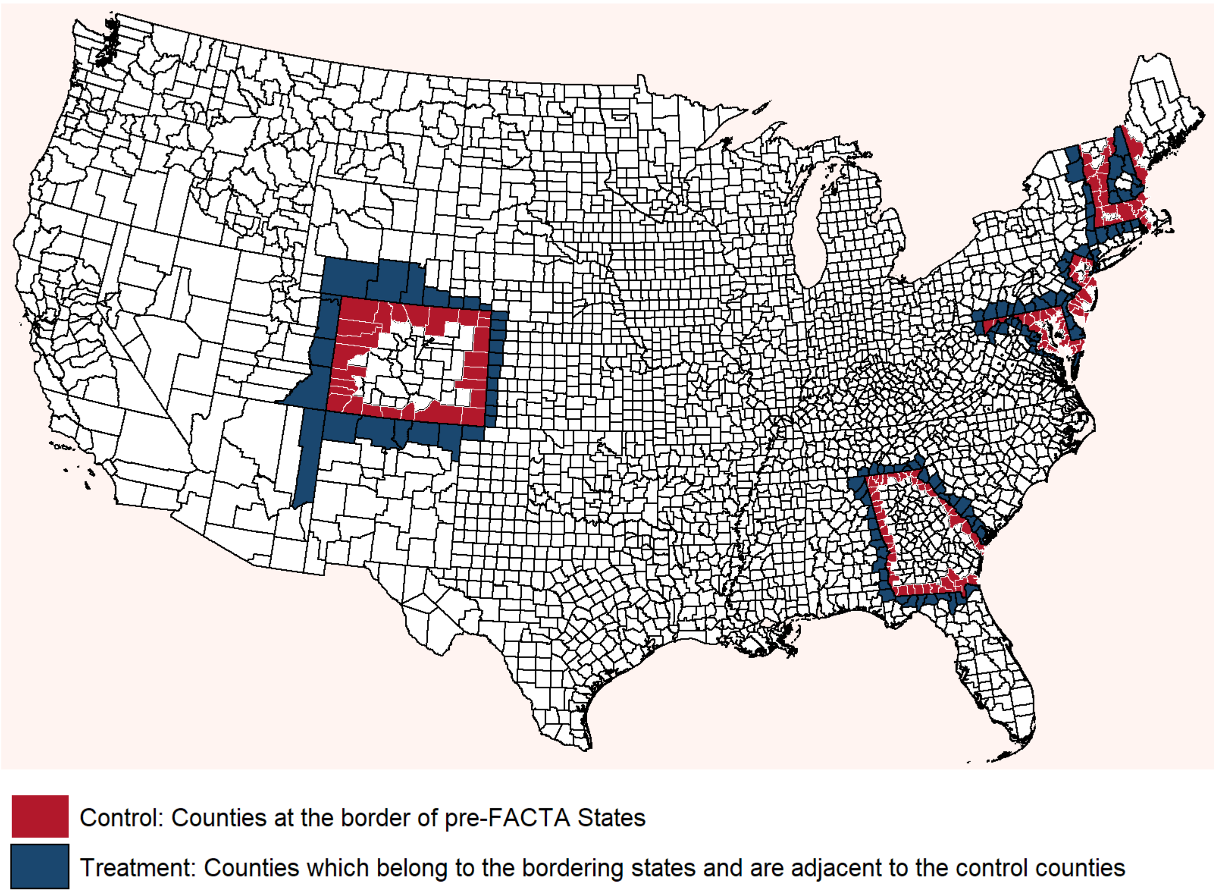

Figure 5:
Difference in Approval Ratio by Years to Treatment

This figure shows the coefficient estimates from regressing "*Approval Ratio*" using the specification:

$$y_{icjt} = \beta_0 + \sum_{k=T-3}^{T-1} \beta_k\, Treament_{ic} \times event_k + \sum_{k=T+1}^{T+4} \beta_k\, Treament_{ic} \times event_k + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$$

where $event_k = 1$ if $t = k$. $event_k = 0$ if $t \neq k$. $T = event\ year$ 2005.

Coefficients are estimated with respect to the base year 2004 $(k = 0)$. The x axis represents time relative to the event year 2005, i.e. T = +1 is the first post-event year period, and so on. The y axis represents the coefficient estimates β_k's and the 95% confidence interval. The regression includes "*Border* × *Year*" fixed effects and "*Census Tract*" fixed effects but no economic controls. Other terms in the equation are the same as those in Equation 1, and are described in detail in Section 3. Standard error is clustered at the county level.

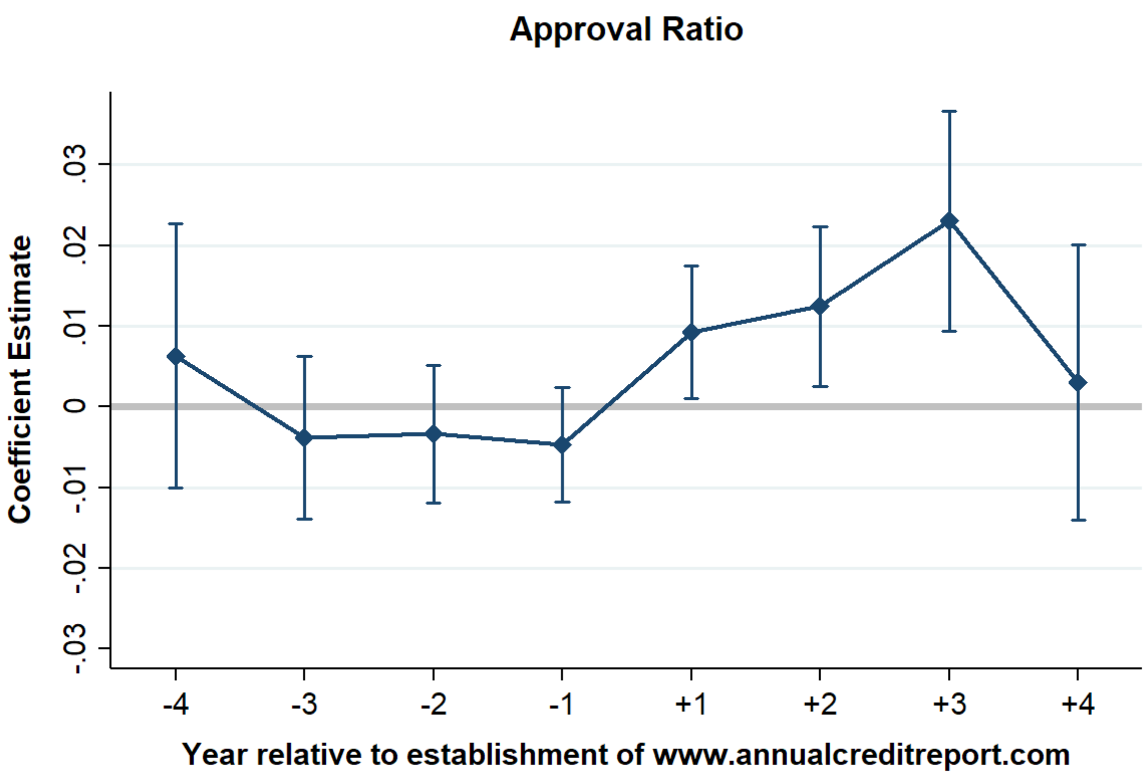

Figure 6:
Google Search Interest for the Term "Free Credit Report" in the U.S.

This figure shows the plot of *Search Interest* for the terms *Free Credit Report* in the US over time from Jan 1, 2004 till Dec 31, 2011. Numbers on the vertical axis represent search interest relative to the highest point on the chart in the US during the time period of the plot. A value of 100 is the peak popularity for the term. A value of 50 means that the term is half as popular. A score of 0 means there was not enough data for this term.

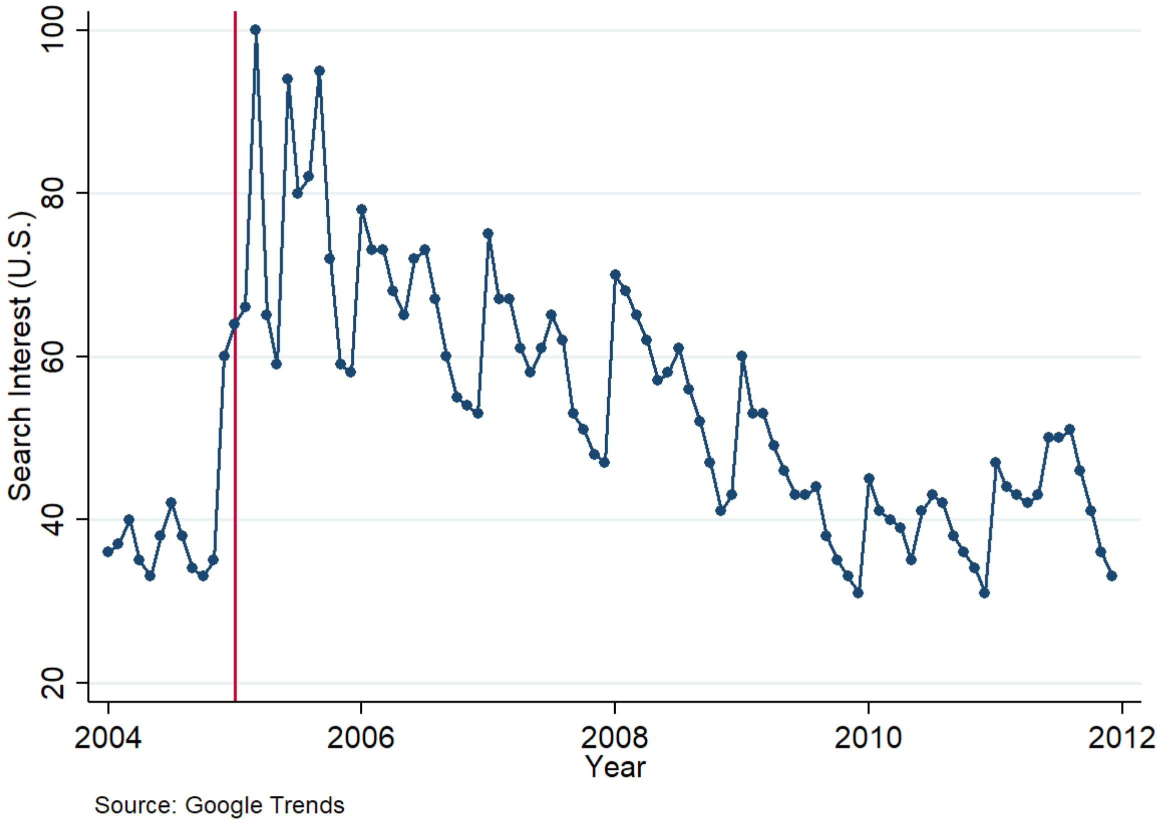

Figure 7:
Popularity of the Keyphrase *Free Credit Report*

This figure shows the difference in mean popularity rank of treatment and control states for the keyphrase *Free Credit Report* from 2004 to 2008. The popularity score of each state is ranges from 0 to 100, calculated each year. A value of 100 represents the location with the highest popularity of the search term as a fraction of total searches in that location. A value of 50 indicates a location where it is half as popular. Here, I plot the difference in mean annual popularity in treatment and control states.

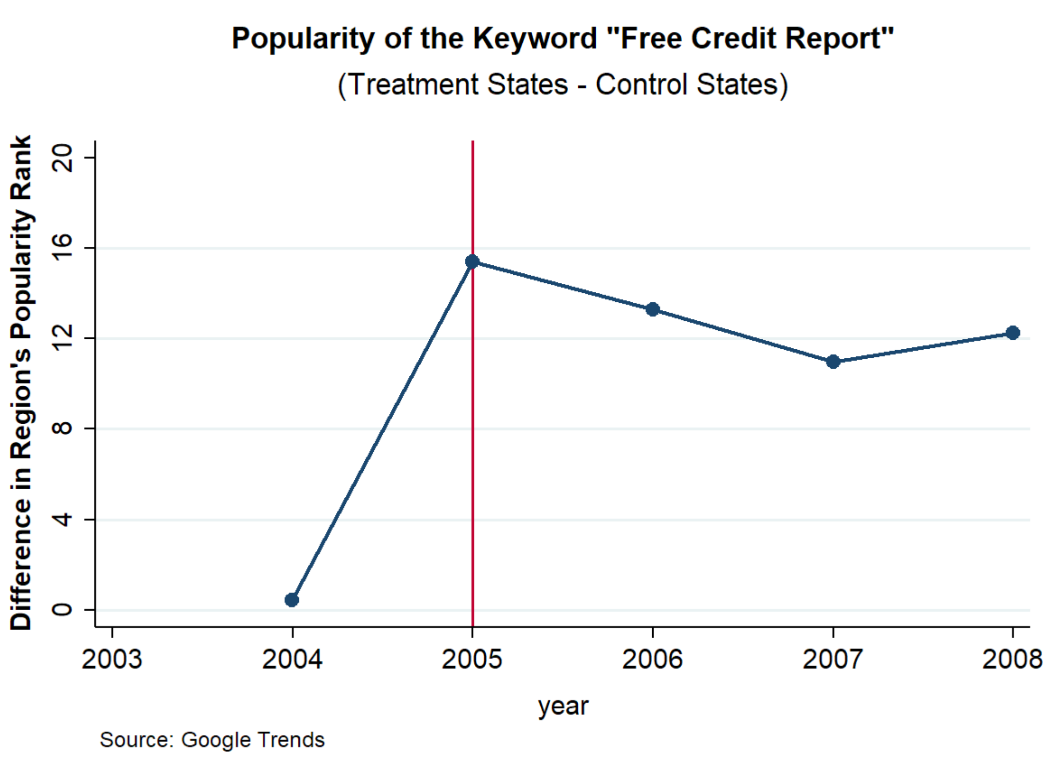

Figure 8:
Free Credit Report and Mortgage Default

This figure shows the adjusted default rate for the sample of 30-years fixed-rate mortgages purchased by Fannie Mae and Freddie Mac. I calculate the percentage of total mortgage originated in the pre-event year 2004 and the post-event year 2005 which goes into default in a month post-origination, $Def_{2005,age}$ and $Def_{2004,age}$, separately for the treated and control zip3-state areas respectively. A mortgage is in default when the scheduled payment is delayed by 30–59 days for the first time. I then calculate the adjusted default rate as:

$$\text{Adjusted Default Rate}_{age} = (Def_{2005,age} - Def_{2004,age})_{treated} - (Def_{2005,age} - Def_{2004,age})_{control}$$

age represent months since origination.

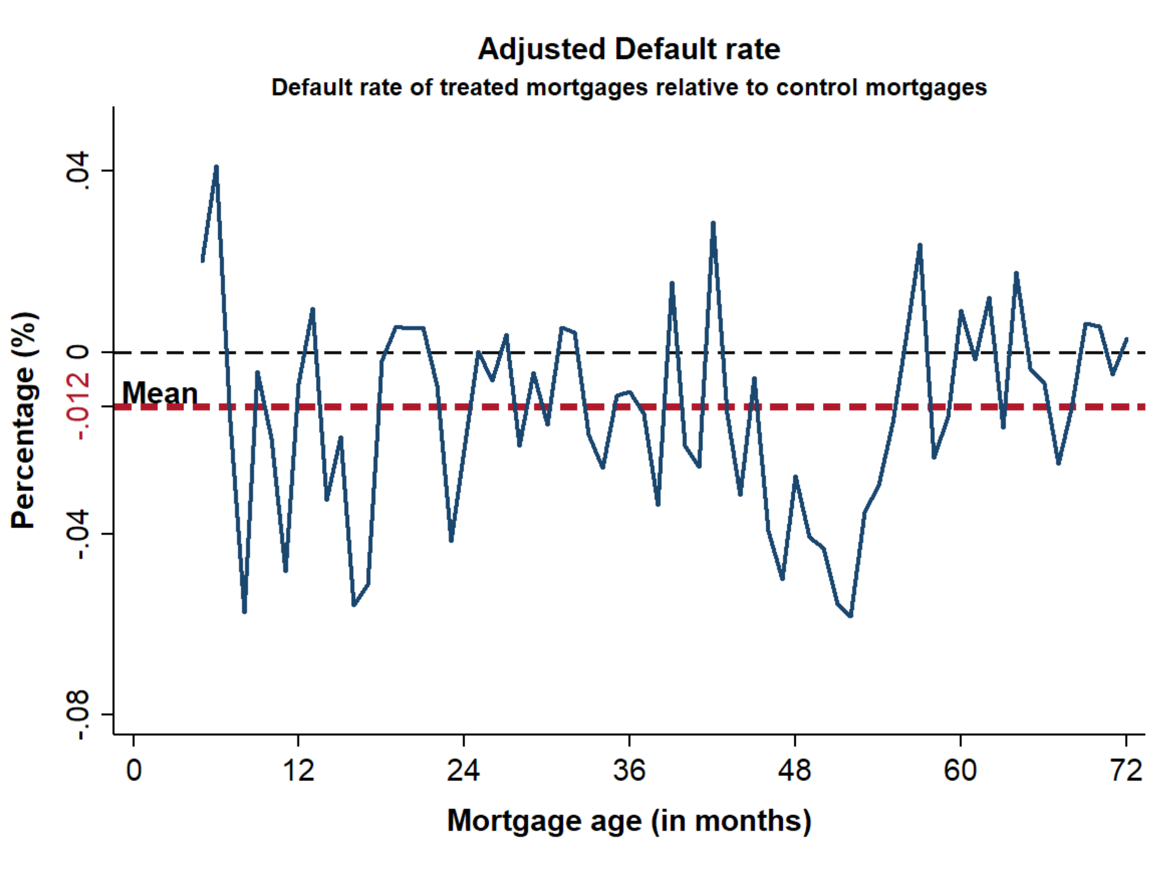

Table 1: Summary Statistics

Panel A shows the statistics for the full sample time period (2000–2008). Panel B shows the statistics are for the pre-treatment period (2000–2004), and the p-values for the t-test for difference in the control and treatment group. *Num of App per 1000 adult* is the number of mortgage applications scaled by the population aged 18 to 64 years in a census tract. *Approval ratio* is the ratio of mortgages originated, or mortgages approved but not accepted to total applications in a census tract. *Deny Credit Hist Ratio* and *Deny Debt-to-inc Ratio* are the ratio of applications denied due to credit history and debt-to-income ratio, respectively, to the number of total applications in a census tract. *Withdrawal Ratio* is the ratio of applications expressly withdrawn by the applicant to the number of total applications in the census tract. The bottom three variables constitute the *Economic controls*. Δ *Inc per capita* is the annual growth rate of county income per capita. Δ *Emp.* is the annual growth rate of the county employment by all establishments; Δ *State GDP* is the annual growth rate of the state gross domestic product.

Panel A: Full Sample (2000 - 2008)

	Full Sample				Control Group (C)				Treatment Group (T)			
	N	Mean	SD	Med.	N	Mean	SD	Med.	N	Mean	SD	Med.
Num of App per 1000 adult	86817	83.42	74.76	66.41	36494	98.42	77.58	77.74	50323	72.54	70.68	56.48
Approval Ratio	82713	0.63	0.13	0.64	35879	0.65	0.12	0.66	46834	0.61	0.13	0.62
Deny Credit Hist Ratio	82713	0.06	0.04	0.05	35879	0.05	0.04	0.04	46834	0.06	0.05	0.05
Deny Debt-to-inc Ratio	82713	0.03	0.03	0.03	35879	0.03	0.02	0.03	46834	0.03	0.03	0.03
Withdrawn Ratio	82713	0.12	0.05	0.12	35879	0.12	0.04	0.11	46834	0.12	0.06	0.12
Δ Inc per capita	88349	0.04	0.03	0.04	37644	0.04	0.03	0.04	50705	0.04	0.03	0.04
Δ Emp	90231	0.01	0.05	0.01	37635	0.01	0.04	0.01	52596	0.01	0.06	0.01
Δ State GDP	90353	0.04	0.02	0.05	37644	0.05	0.02	0.04	52709	0.04	0.02	0.05

Panel B: Pre - Treatment Sample (2000 - 2004)

	Full Sample				Control Group (C)				Treatment Group (T)				(C-T)
	N	Mean	SD	Med.	N	Mean	SD	Med.	N	Mean	SD	Med.	p-val
Num of App per 1000 adult	48367	110.52	83.57	93.50	20291	129.70	86.15	108.66	28076	96.65	78.79	83.22	0.000
Approval Ratio	47028	0.64	0.13	0.65	20074	0.67	0.12	0.68	26954	0.62	0.13	0.63	0.000
Deny Credit Hist Ratio	47028	0.06	0.04	0.05	20074	0.06	0.04	0.05	26954	0.07	0.05	0.06	0.000
Deny Debt-to-inc Ratio	47028	0.03	0.02	0.03	20074	0.03	0.02	0.03	26954	0.03	0.02	0.03	0.000
Withdrawn Ratio	47028	0.12	0.05	0.11	20074	0.12	0.04	0.11	26954	0.13	0.05	0.12	0.000
Δ Inc per capita	50293	0.04	0.03	0.04	21372	0.04	0.03	0.04	28921	0.04	0.03	0.04	0.000
Δ Emp	51342	0.01	0.06	0.01	21366	0.01	0.04	0.01	29976	0.01	0.07	0.01	0.000
Δ State GDP	51452	0.05	0.02	0.05	21372	0.05	0.02	0.04	30080	0.04	0.02	0.05	0.000

Table 2: Mortgage Applications and the Mortgage Approval Ratio

This table reports the estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio. The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). *#Applications* and *Approval Ratio* are the number of applications per 1000 adult and the mortgage approval ratio in a census tract, respectively. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include the $Border \times Year$ fixed effects (FE) and the $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)
	# Applications	# Applications	Approval Ratio	Approval Ratio
Treat × Post	13.29***	15.45***	0.01***	0.01***
	(2.94)	(3.65)	(2.73)	(2.97)
Economic controls	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.807	0.808	0.739	0.735
Observations	86807	84786	82667	80665

Table 3: Owner-Occupied Mortgages

Panel A of this table reports the estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio only for the sub-sample of owner-occupied mortgage applications. Panel B reports the result of regressing the ratio of not owner-occupied mortgages to total applications (to successful applications) in column 1 and 2 (in column 3 and 4). The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1).

#Applications and *Approval Ratio* are the number of applications per 1000 adults and the mortgage approval ratio in a census tract, respectively. *Non-ocp.* represents the fraction of total (successful) applications which are not owner occupied in Column 1 and 2 (3 and 4). The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and *Census Tract* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Applications and Approval Ratio for Owner Occupied Mortgages

	(1)	(2)	(3)	(4)
	# Applications	# Applications	Approval Ratio	Approval Ratio
Treat × Post	9.81***	11.50***	0.01***	0.01***
	(2.83)	(3.48)	(2.89)	(3.04)
Economic controls	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.640	0.645	0.618	0.613
Observations	86807	84786	84190	82256

Panel B: Fraction of Owner Occupied Mortgages

	Fraction of total app.		Fraction of successful app.	
	(1)	(2)	(3)	(4)
	Non-ocp.	Non-ocp.	Non-ocp.	Non-ocp.
Treat × Post	0.01	0.01	0.01	0.01*
	(1.55)	(1.60)	(1.58)	(1.69)
Economic controls	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.129	0.127	0.105	0.104
Observations	82667	80665	82626	80624

Table 4: Credit History Improvement

This table reports the estimates of the treatment effect on the fraction of mortgage applications denied because of credit history and debt-to-income ratio, estimated separately. The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1).

$\%C.Hist$ ($\%DTI$) is calculated as the ratio of the number of denied applications due to credit history (debt-to-income ratio) to the total number of mortgage applications in a census tract. *High Denial Areas* are the census tracts where denial per capita in the pre-event year 2004 was more than the regional mean of denials across the census tracts. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the fraction of mortgage applications denied due to a given reason in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and *Census Tract* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	All Areas		High Denial Areas		All Areas		High Denial Areas	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	% C.Hist	% C.Hist	% C.Hist	% C.Hist	% DTI	% DTI	% DTI	% DTI
Treat × Post	-0.003	-0.003	-0.003**	-0.003*	-0.002	-0.001	-0.002	-0.002
	(-1.50)	(-1.51)	(-2.04)	(-1.88)	(-1.07)	(-0.96)	(-1.48)	(-1.19)
Economic Controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.542	0.538	0.575	0.575	0.267	0.266	0.319	0.322
Observations	82667	80665	39071	38692	82667	80665	39071	38692

Table 5: Credit Shopping

This table reports the estimates of the treatment effect on the fraction of mortgage applications withdrawn. The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). *% Application Withdrawn* is the ratio of applications expressly withdrawn by consumers to the number of applications in a census tract. The coefficient associated with the $Treat \times Post$ interaction term captures the change in fraction of applications expressly withdrawn by applicants in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and *Census Tract* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)
	% Application Withdrawn	% Application Withdrawn
Treat × Post	-0.009***	-0.011***
	(-2.81)	(-3.51)
Economic controls	No	Yes
Census Tract FE	Yes	Yes
Border × Year FE	Yes	Yes
Cluster (County)	Yes	Yes
R^2 (Adj.)	0.340	0.341
Observations	82667	80665

Table 6: First-time Homebuyers

This table reports the estimates of the treatment effect on the fraction of first-time homebuyers. The regression specification is:

$y_{zcjt} = \beta_0 + \beta_1 Treament_{zc} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (2). The mortgage data used in this table is from the Fannie Mae and Freddie Mac combined single-family loan dataset (GSE data). The dependent variable in column 1 (2) is the ratio of the number of first-time homebuyers to the total number of mortgages (total number of mortgages for which the information on the first-time homebuyer is not missing) in a given zip3-state area. The coefficient associated with the $Treat \times Post$ interaction term captures the change in proportion of *First-time homebuyers* in the treated 3-digit zipcode-state areas relative to the control 3 digit zipcode areas. All regressions include $Border \times Quarter$ fixed effects (FE) and $Zip3\text{-}State$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Denominator - Applications with Known Status		Denominator - All Applications	
	(1)	(2)	(3)	(4)
	% First-time	% First-time	% First-time	% First-time
Treat × Post	0.01**	0.01**	0.01**	0.01*
	(2.55)	(2.31)	(2.00)	(1.78)
Economic Controls	No	Yes	No	Yes
Zip3-State FE	Yes	Yes	Yes	Yes
Border × Qtr FE	Yes	Yes	Yes	Yes
Cluster Zip3-State	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.691	0.692	0.691	0.691
Observations	7706	7706	7711	7711

Table 7: Effect of Free Credit Report: Role of Creditworthiness (County Measure) This table reports estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio separately for prime and subprime counties. A county is categorized as subprime if its subprime population fraction is more than the *regional mean* subprime population fraction. The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). *#App.* and *Apv. Ratio* are the number of applications per 1000 adults and the mortgage approval ratio in a census tract, respectively. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Prime Counties				Subprime Counties			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	# App.	# App.	Apv. Ratio	Apv. Ratio	# App.	# App.	Apv. Ratio	Apv. Ratio
Treat × Post	16.73**	18.11**	0.02***	0.02***	8.49	10.69**	0.01*	0.01**
	(2.29)	(2.59)	(3.14)	(3.32)	(1.64)	(2.25)	(1.70)	(2.01)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.802	0.802	0.777	0.773	0.826	0.828	0.681	0.680
Observations	39263	37698	38184	36631	47241	46797	44375	43938

Table 8: Effect of Free Credit Report: Role of Creditworthiness (Census Tract Measure)

This table reports estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio separately for census tracts with low and high density of payday lenders. The regression specification is: $y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). #*App.* and *Apv. Ratio* are the number of applications per 1000 adults and the mortgage approval ratio in a census tract, respectively. The benchmark for creating the high and low number of payday lenders areas is the average number of payday lenders in census tracts in counties at the border of Colorado (CO) and surrounding states in 2004. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	# Payday Lenders - Low				# Payday Lenders - High			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	# App.	# App.	Apv. Ratio	Apv. Ratio	# App.	# App.	Apv. Ratio	Apv. Ratio
Treat × Post	68.43***	72.78***	0.05***	0.05***	43.72***	43.27***	0.01	0.01
	(5.31)	(5.36)	(3.46)	(3.43)	(3.82)	(4.00)	(0.60)	(0.64)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.790	0.793	0.751	0.751	0.816	0.818	0.773	0.774
Observations	1452	1452	1395	1395	872	872	865	865

Table 9: Effect of Free Credit Report: Role of Education

This table reports estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio separately for the census tracts with a high and low graduate fraction population. The cut-off fraction for defining the high and low graduate fraction population is the *regional mean* graduate population fraction (see Footnote 29). The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1).
#App. and *Apv. Ratio* are the number of applications per 1000 adults and the mortgage approval ratio in a census tract, respectively. Graduate education is defined as *associate degree*, *bachelor's degree*, or *graduate or professional degree*. The graduate fraction population is calculated as the number of adults aged 18 to 64 having graduate degree to the total number of adults aged 18 to 64 in a given census tract, as recorded in Census 2000. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Low Education Area				High Education Area			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	# App.	# App.	Apv. Ratio	Apv. Ratio	# App.	# App.	Apv. Ratio	Apv. Ratio
Treat × Post	6.03*	6.88**	0.01	0.01*	12.52***	15.04***	0.01***	0.01***
	(1.95)	(2.20)	(1.55)	(1.66)	(2.67)	(3.55)	(2.92)	(3.08)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.828	0.831	0.582	0.575	0.824	0.823	0.743	0.739
Observations	35945	35419	33182	32666	50720	49261	49363	47913

Table 10: Effect of Free Credit Report: Role of Income

This table reports estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio for each income quartile. The regression specification is:

$$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}. \quad \text{See Eq. (1).}$$

#App. and *Apv. Ratio* are the number of applications per 1000 adults and the mortgage approval ratio in a census tract, respectively. Income quartiles are calculated every year for a given census tract. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include *Border* $\times$ *Year* fixed effects (FE) and *Census Tract* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Income Quartile 1 and 2

	Income Quartile 1				Income quartile 2			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	# App.	# App.	Apv. Ratio	Apv. Ratio	# App.	# App.	Apv. Ratio	Apv. Ratio
Treat × Post	-0.25	-0.41	0.01***	0.02***	2.44**	2.67***	0.01	0.01
	(-0.18)	(-0.29)	(2.99)	(3.97)	(2.47)	(2.68)	(1.03)	(1.39)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.779	0.780	0.257	0.259	0.802	0.804	0.340	0.339
Observations	88283	86252	79920	77983	88283	86252	81114	79136

Panel B: Income Quartile 3 and 4

	Income Quartile 3				Income quartile 4			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	# App.	# App.	Apv. Ratio	Apv. Ratio	# App.	# App.	Apv. Ratio	Apv. Ratio
Treat × Post	3.30***	3.88***	0.01	0.01	6.10**	7.50***	0.01*	0.01**
	(2.77)	(3.55)	(1.13)	(1.23)	(2.20)	(3.08)	(1.87)	(2.06)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.795	0.797	0.399	0.394	0.739	0.739	0.379	0.373
Observations	88283	86252	81634	79639	88283	86252	81337	79339

Table 11: Effect of Free Credit Report on House Prices

This table reports estimates of the treatment effect of free credit reports on the house price and growth in house price. The regression specification is:

$$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}. \quad \text{See Eq. (1).}$$

Index and Δ *Index* are the house price index and change in house price index $(Index_t - Index_{t-1})$, respectively. Value of the house price index is 100 in year 2000. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)
	Index	Index	ΔIndex	ΔIndex
Treat × Post	0.89	0.89	2.38*	2.44*
	(0.21)	(0.21)	(1.77)	(1.70)
Economic controls	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.934	0.934	0.676	0.678
Observations	28572	28546	25387	25361

Table 12: Disentangling the Supply and Demand Effect using Interest Rate

This table reports the estimates of the treatment effect of free credit reports on mean and median interest rate in a zip3-state area. The regression specification is:

$$y_{cjt} = \beta_0 + \beta_1 Treament_c \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{ct}. \quad \text{See Eq. (\textbf{??}).}$$

Mean (*Median*) *Int Rate* is the mean (median) interest rate expressed in percentage points, calculated in a zip3-state area. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and *County* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Interest Rate (in percentage point)			
	(1)	(2)	(3)	(4)
	Mean	Mean	Median	Median
Treat × Post	0.012*	0.011*	0.011*	0.010*
	(1.93)	(1.77)	(1.93)	(1.72)
Economic Controls	No	Yes	No	Yes
Zip3-State FE	Yes	Yes	Yes	Yes
Border × Qtr FE	Yes	Yes	Yes	Yes
Cluster Zip3-State	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.991	0.991	0.989	0.989
Observations	7711	7711	7711	7711

Table 13: Disentangling the Supply Effect - Heterogeneity in Lender Density

This table reports the estimates of the treatment effect on the origination volume (in 1000 USD) per adult and the mortgage approval ratio, estimated separately for census tracts having a high and low density of mortgage lenders per capita in 2004. The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). *Low* (*High*) identifies a census tract having a lower (higher) number of HMDA lenders than the *regional mean* number of HMDA lenders (per census tract) within the bordering counties between the given control state and all the treatment states surrounding it in 2004 (See Footnote 29). *Difference* [*High* - *Low*] shows the result for the t-test for the difference in coefficients of $Treat \times Post$ in specifications *High* and *Low*. The dependent variable in columns 1 through 4 is volume of mortgage originated (in 1000 USD) per adult in a census tract. The dependent variable in columns 4 through 8 is the mortgage approval ratio of mortgage applications at census tract-level. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Volume (in 1000 USD) per Adult				Approval Ratio			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low	High	Low	High	Low	High	Low	High
Treat × Post	0.002**	0.001	0.003***	0.002*	0.015***	0.009*	0.015***	0.010**
	(2.16)	(1.11)	(2.84)	(1.78)	(2.98)	(1.90)	(3.21)	(2.03)
Difference [High - Low]		-0.001		-0.001		-0.006		-0.006
p-value		(0.577)		(0.581)		(0.501)		(0.475)
Economic Controls	No	No	Yes	Yes	No	No	Yes	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.663	0.595	0.652	0.589	0.750	0.716	0.746	0.712
Observations	60699	25805	59206	25289	57623	24936	56143	24426

Table 14: Did Approval Ratio Increase due to Private Securitization?

This table reports the estimates of the treatment effect on the mortgage approval ratio estimated separately for mortgages not sold, sold to government agencies (GSEs), and mortgages sold to others (non-GSE). The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1).

The dependent variables are: the fraction of total mortgage applications originated and sold to the non-GSEs (columns (1) and (2)); originated and sold to the GSEs (columns (3) and (4)); and approved and not sold by the lending institutions (columns (5) and (6)). The dependent variables are calculated at the census tract-level. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Sold to Non-GSE		Sold to GSE		Not Sold	
	(1)	(2)	(3)	(4)	(5)	(6)
	Fraction	Fraction	Fraction	Fraction	Fraction	Fraction
Treat × Post	-0.007	-0.006	0.039**	0.042**	-0.004	-0.001
	(-0.54)	(-0.48)	(2.22)	(2.33)	(-0.58)	(-0.12)
Economic controls	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.003	0.002	-0.000	-0.000	0.032	0.034
Observations	82667	80665	82667	80665	82667	80665

Table 15: Prime Borrowers - Credit Score-Based Evidence

This table reports the estimates of the treatment effect on the number of mortgages originated to prime and subprime borrowers by government sponsored enterprises (GSE's) – Fannie Mae and Freddie Mac. The regression specification is:

$y_{zcjt} = \beta_0 + \beta_1 Treament_{zc} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (2).

The dependent variable in column 1 is *Number of Origination to Prime Borrowers* (credit score$\geq$620) in a given zip3-state area. The dependent variable in column 2 is *Number of Applications to Subprime Borrowers* (credit score<620) in a given zip3-state area. The coefficient associated with the *Treat* $\times$ *Post* interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include *Border* $\times$ *Year* fixed effects (FE) and *Census Tract* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	(1) #Prime App.	(2) #Prime App.	(3) #Subprime App.	(4) #Subprime App.
Treat $\times$ Post	308.58***	312.51***	10.48**	10.78**
	(3.39)	(3.33)	(2.12)	(2.16)
Economic Controls	No	Yes	No	Yes
Zip3-State FE	Yes	Yes	Yes	Yes
Border $\times$ Qtr FE	Yes	Yes	Yes	Yes
Cluster Zip3-State	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.757	0.758	0.792	0.792
Observations	7711	7711	7711	7711

A Appendix

For robustness, I re-estimate all the tables excluding the years 2007 and 2008. The results are provided in this appendix.

Table A.2: Mortgage Applications and the Mortgage Approval Ratio

This table reports the estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio. The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). *#Applications* and *Approval Ratio* are the number of applications per 1000 adult and the mortgage approval ratio in a census tract, respectively. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include the $Border \times Year$ fixed effects (FE) and the $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)
	# Applications	# Applications	Approval Ratio	Approval Ratio
Treat × Post	10.18**	9.68***	0.01**	0.01***
	(2.43)	(2.77)	(2.46)	(2.66)
Economic controls	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.811	0.813	0.780	0.775
Observations	67603	66016	65068	63493

Table A.3: Owner-Occupied Mortgages

Panel A of this table reports the estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio only for the sub-sample of owner-occupied mortgage applications. Panel B reports the result of regressing the ratio of not owner-occupied mortgages to total applications (to successful applications) in column 1 and 2 (in column 3 and 4). The regression specification is:

$$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}. \quad \text{See Eq. (1).}$$

#Applications and *Approval Ratio* are the number of applications per 1000 adults and the mortgage approval ratio in a census tract, respectively. *Non-ocp.* represents the fraction of total (successful) applications which are not owner occupied in Column 1 and 2 (3 and 4). The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Applications and Approval Ratio for Owner Occupied Mortgages

	(1)	(2)	(3)	(4)
	# Applications	# Applications	Approval Ratio	Approval Ratio
Treat × Post	6.84**	7.08**	0.01**	0.01**
	(2.00)	(2.17)	(2.49)	(2.53)
Economic controls	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.634	0.639	0.665	0.660
Observations	67603	66016	65056	63556

Panel B: Fraction of Owner Occupied Mortgages

	Fraction of total app.		Fraction of successful app.	
	(1)	(2)	(3)	(4)
	Non-ocp.	Non-ocp.	Non-ocp.	Non-ocp.
Treat × Post	0.01	0.01*	0.01	0.01*
	(1.60)	(1.96)	(1.54)	(1.94)
Economic controls	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.093	0.092	0.075	0.074
Observations	65068	63493	65041	63466

Table A.4: Credit History Improvement

This table reports the estimates of the treatment effect on the fraction of mortgage applications denied because of credit history and debt-to-income ratio, estimated separately. The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). $\%C.Hist$ ($\%DTI$) is calculated as the ratio of the number of denied applications due to credit history (debt-to-income ratio) to the total number of mortgage applications in a census tract. *High Denial Areas* are the census tracts where denial per capita in the pre-event year 2004 was more than the regional mean of denials across the census tracts. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the fraction of mortgage applications denied due to a given reason in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and *Census Tract* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	All Areas		High Denial Areas		All Areas		High Denial Areas	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	% C.Hist	% C.Hist	% C.Hist	% C.Hist	% DTI	% DTI	% DTI	% DTI
Treat × Post	-0.003	-0.004*	-0.004*	-0.004*	-0.002**	-0.003***	-0.003***	-0.003***
	(-1.55)	(-1.93)	(-1.89)	(-1.97)	(-2.39)	(-3.25)	(-2.91)	(-3.14)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.596	0.593	0.643	0.643	0.259	0.258	0.322	0.322
Observations	65068	63493	30455	30153	65068	63493	30455	30153

Table A.5: Credit Shopping

This table reports the estimates of the treatment effect on the fraction of mortgage applications withdrawn. The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). *% Application Withdrawn* is the ratio of applications expressly withdrawn by consumers to the number of applications in a census tract. The coefficient associated with the *Treat* × *Post* interaction term captures the change in fraction of applications expressly withdrawn by applicants in the treated census tracts relative to the control census tracts. All regressions include *Border* × *Year* fixed effects (FE) and *Census Tract* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)
	% Application Withdrawn	% Application Withdrawn
Treat × Post	-0.007**	-0.008***
	(-2.53)	(-2.94)
Economic controls	No	Yes
Census Tract FE	Yes	Yes
Border × Year FE	Yes	Yes
Cluster (County)	Yes	Yes
R^2 (Adj.)	0.403	0.399
Observations	65068	63493

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.6: First-time Homebuyers

This table reports the estimates of the treatment effect on the fraction of first-time homebuyers. The regression specification is:

$y_{zcjt} = \beta_0 + \beta_1 Treament_{zc} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (2). The mortgage data used in this table is from the Fannie Mae and Freddie Mac combined single-family loan dataset (GSE data). The dependent variable in column 1 (2) is the ratio of the number of first-time homebuyers to the total number of mortgages (total number of mortgages for which the information on the first-time homebuyer is not missing) in a given zip3-state area. The coefficient associated with the $Treat \times Post$ interaction term captures the change in proportion of *First-time homebuyers* in the treated 3-digit zipcode-state areas relative to the control 3 digit zipcode areas. All regressions include $Border \times Quarter$ fixed effects (FE) and $Zip3\text{-}State$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Denominator - Applications with Known Status		Denominator - All Applications	
	(1)	(2)	(3)	(4)
	% First-time	% First-time	% First-time	% First-time
Treat × Post	0.01***	0.01**	0.01**	0.01*
	(2.77)	(2.35)	(2.27)	(1.87)
Economic Controls	No	Yes	No	Yes
Zip3-State FE	Yes	Yes	Yes	Yes
Border × Qtr FE	Yes	Yes	Yes	Yes
Cluster Zip3-State	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.674	0.674	0.671	0.671
Observations	6014	6014	6017	6017

Table A.7: Effect of Free Credit Report: Role of Creditworthiness (County Measure)

This table reports estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio separately for prime and subprime counties. A county is categorized as subprime if its subprime population fraction is more than the *regional mean* subprime population fraction. The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1).

#App. and *Apv. Ratio* are the number of applications per 1000 adults and the mortgage approval ratio in a census tract, respectively. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Prime Counties				Subprime Counties			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	# App.	# App.	Apv. Ratio	Apv. Ratio	# App.	# App.	Apv. Ratio	Apv. Ratio
Treat × Post	14.61**	14.73**	0.01***	0.01***	5.38	5.02	0.01**	0.02**
	(2.21)	(2.44)	(3.23)	(3.78)	(1.25)	(1.29)	(2.23)	(2.50)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.800	0.802	0.814	0.809	0.835	0.836	0.729	0.727
Observations	30552	29315	29857	28628	36786	36448	35106	34772

Table A.8: Effect of Free Credit Report: Role of Creditworthiness (Census Tract Measure)

This table reports estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio separately for census tracts with low and high density of payday lenders. The regression specification is: $y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). #*App.* and *Apv. Ratio* are the number of applications per 1000 adults and the mortgage approval ratio in a census tract, respectively. The benchmark for creating the high and low number of payday lenders areas is the average number of payday lenders in census tracts in counties at the border of Colorado (CO) and surrounding states in 2004. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	# Payday Lenders - Low				# Payday Lenders - High			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	# App.	# App.	Apv. Ratio	Apv. Ratio	# App.	# App.	Apv. Ratio	Apv. Ratio
Treat × Post	59.04***	62.64***	0.05***	0.05***	40.05***	40.05***	0.03	0.02
	(5.05)	(5.32)	(3.48)	(3.11)	(4.35)	(4.39)	(1.49)	(1.35)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.790	0.794	0.799	0.798	0.808	0.813	0.820	0.821
Observations	1131	1131	1095	1095	679	679	676	676

Table A.9: Effect of Free Credit Report: Role of Education

This table reports estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio separately for the census tracts with a high and low graduate fraction population. The cut-off fraction for defining the high and low graduate fraction population is the *regional mean* graduate population fraction (see Footnote 29). The regression specification is: $y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). *#App.* and *Apv. Ratio* are the number of applications per 1000 adults and the mortgage approval ratio in a census tract, respectively. Graduate education is defined as *associate degree*, *bachelor's degree*, or *graduate or professional degree*. The graduate fraction population is calculated as the number of adults aged 18 to 64 having graduate degree to the total number of adults aged 18 to 64 in a given census tract, as recorded in Census 2000. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Low Education Area				High Education Area			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	# App.	# App.	Apv. Ratio	Apv. Ratio	# App.	# App.	Apv. Ratio	Apv. Ratio
Treat × Post	3.45	3.54	0.01	0.01	9.74**	9.47***	0.01***	0.01***
	(1.22)	(1.27)	(0.92)	(1.12)	(2.22)	(2.76)	(2.83)	(3.00)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.840	0.843	0.639	0.631	0.824	0.824	0.783	0.778
Observations	28027	27612	26334	25926	39472	38326	38644	37503

Table A.10: Effect of Free Credit Report: Role of Income

This table reports estimates of the treatment effect of free credit reports on the number of mortgage applications and the mortgage approval ratio for each income quartile. The regression specification is:

$$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}. \quad \text{See Eq. (1).}$$

#App. and *Apv. Ratio* are the number of applications per 1000 adults and the mortgage approval ratio in a census tract, respectively. Income quartiles are calculated every year for a given census tract. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and $Census\ Tract$ FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Income Quartile 1 and 2

	Income Quartile 1				Income quartile 2			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	# App.	# App.	Apv. Ratio	Apv. Ratio	# App.	# App.	Apv. Ratio	Apv. Ratio
Treat × Post	-0.38	-1.06	0.01**	0.02***	1.69**	1.48*	0.00	0.01
	(-0.29)	(-0.90)	(2.52)	(3.29)	(1.99)	(1.91)	(0.50)	(1.04)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.798	0.801	0.295	0.297	0.815	0.818	0.386	0.383
Observations	68869	67272	63516	61969	68869	67272	64154	62590

Panel B: Income Quartile 3 and 4

	Income Quartile 3				Income quartile 4			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	# App.	# App.	Apv. Ratio	Apv. Ratio	# App.	# App.	Apv. Ratio	Apv. Ratio
Treat × Post	2.13*	2.03**	0.01	0.01	5.13*	5.54***	0.01	0.01*
	(1.93)	(2.08)	(0.89)	(1.09)	(1.95)	(2.62)	(1.47)	(1.72)
Economic controls	No	Yes	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.811	0.814	0.436	0.430	0.760	0.761	0.415	0.408
Observations	68869	67272	64395	62822	68869	67272	64216	62641

Table A.11: Effect of Free Credit Report on House Prices

This table reports estimates of the treatment effect of free credit reports on the house price and growth in house price. The regression specification is:

$$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}. \quad \text{See Eq. (1).}$$

Index and Δ *Index* are the house price index and change in house price index $(Index_t - Index_{t-1})$, respectively. Value of the house price index is 100 in year 2000. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and *Census Tract* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)
	Index	Index	ΔIndex	ΔIndex
Treat × Post	-1.37	-1.48	1.39	1.29
	(-0.35)	(-0.38)	(0.75)	(0.73)
Economic controls	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.938	0.938	0.619	0.621
Observations	22231	22205	19053	19027

Table A.12: Disentangling the Supply and Demand Effect using Interest Rate

This table reports the estimates of the treatment effect of free credit reports on mean and median interest rate in a zip3-state area. The regression specification is: $y_{cjt} = \beta_0 + \beta_1 Treament_c \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{ct}$. See Eq. (**??**). *Mean* (*Median*) *Int Rate* is the mean (median) interest rate expressed in percentage points, calculated in a zip3-state area. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include $Border \times Year$ fixed effects (FE) and *County* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Interest Rate (in percentage point)			
	(1)	(2)	(3)	(4)
	Mean	Mean	Median	Median
Treat × Post	0.01*	0.01*	0.01*	0.01*
	(1.84)	(1.70)	(1.72)	(1.84)
Economic Controls	No	Yes	No	Yes
Zip3-State FE	Yes	Yes	Yes	Yes
Border × Qtr FE	Yes	Yes	Yes	Yes
Cluster Zip3-State	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.993	0.993	0.991	0.991
Observations	6017	6017	6017	6017

Table A.13: Disentangling the Supply Effect - Heterogeneity in Lender Density

This table reports the estimates of the treatment effect on the origination volume (in 1000 USD) per adult and the mortgage approval ratio, estimated separately for census tracts having a high and low density of mortgage lenders per capita in 2004. The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1). *Low* (*High*) identifies a census tract having a lower (higher) number of HMDA lenders than the *regional mean* number of HMDA lenders (per census tract) within the bordering counties between the given control state and all the treatment states surrounding it in 2004 (See Footnote 29). *Difference* [*High* - *Low*] shows the result for the t-test for the difference in coefficients of $Treat \times Post$ in specifications *High* and *Low*. The dependent variable in columns 1 through 4 is volume of mortgage originated (in 1000 USD) per adult in a census tract. The dependent variable in columns 4 through 8 is the mortgage approval ratio of mortgage applications at census tract-level. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Volume (in 1000 USD) per Adult				Approval Ratio			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low	High	Low	High	Low	High	Low	High
Treat × Post	0.002*	0.001	0.002*	0.001	0.013**	0.010**	0.014**	0.012**
	(1.76)	(0.83)	(1.96)	(1.05)	(2.47)	(2.14)	(2.57)	(2.49)
Difference [High - Low]		-0.001***		-0.001***		-0.003***		-0.002***
p-value		(0.000)		(0.000)		(0.000)		(0.000)
Economic Controls	No	No	Yes	Yes	No	No	Yes	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.672	0.584	0.662	0.578	0.790	0.761	0.785	0.756
Observations	47264	20074	46095	19668	45368	19595	44207	19193

Table A.14: Did Approval Ratio Increase due to Private Securitization?
This table reports the estimates of the treatment effect on the mortgage approval ratio estimated separately for mortgages not sold, sold to government agencies (GSEs), and mortgages sold to others (non-GSE). The regression specification is:

$y_{icjt} = \beta_0 + \beta_1 Treament_{ic} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (1).

The dependent variables are: the fraction of total mortgage applications originated and sold to the non-GSEs (columns (1) and (2)); originated and sold to the GSEs (columns (3) and (4)); and approved and not sold by the lending institutions (columns (5) and (6)). The dependent variables are calculated at the census tract-level. The coefficient associated with the $Treat \times Post$ interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Sold to Non-GSE		Sold to GSE		Not Sold	
	(1)	(2)	(3)	(4)	(5)	(6)
	Fraction	Fraction	Fraction	Fraction	Fraction	Fraction
Treat × Post	-0.006	-0.005	0.042**	0.042***	-0.007	-0.003
	(-0.53)	(-0.49)	(2.32)	(2.65)	(-0.99)	(-0.49)
Economic controls	No	Yes	No	Yes	No	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes
Border × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster (County)	Yes	Yes	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.003	0.002	-0.000	-0.001	0.041	0.024
Observations	65068	63493	65068	63493	65068	63493

Table A.15: Prime Borrowers - Credit Score-Based Evidence

This table reports the estimates of the treatment effect on the number of mortgages originated to prime and subprime borrowers by government sponsored enterprises (GSE's) – Fannie Mae and Freddie Mac. The regression specification is:

$y_{zcjt} = \beta_0 + \beta_1 Treament_{zc} \times Post_T + \delta \times Economic\_controls + \alpha_i + \gamma_{j,t} + \varepsilon_{it}$. See Eq. (2). The dependent variable in column 1 is *Number of Origination to Prime Borrowers* (credit score>620) in a given zip3-state area. The dependent variable in column 2 is *Number of Applications to Subprime Borrowers* (credit score≤620) in a given zip3-state area. The coefficient associated with the *Treat* × *Post* interaction term captures the change in the dependent variable in the treated census tracts relative to the control census tracts. All regressions include *Border* × *Year* fixed effects (FE) and *Census Tract* FE. All variables are defined in Table 1. Standard errors are clustered by county. t-statistics are reported below the coefficients in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)
	#Prime App.	#Prime App.	#Subprime App.	#Subprime App.
Treat × Post	289.56***	301.41***	6.54	6.66
	(3.18)	(3.24)	(1.37)	(1.39)
Economic Controls	No	Yes	No	Yes
Zip3-State FE	Yes	Yes	Yes	Yes
Border × Qtr FE	Yes	Yes	Yes	Yes
Cluster Zip3-State	Yes	Yes	Yes	Yes
R^2 (Adj.)	0.761	0.762	0.827	0.827
Observations	6017	6017	6017	6017